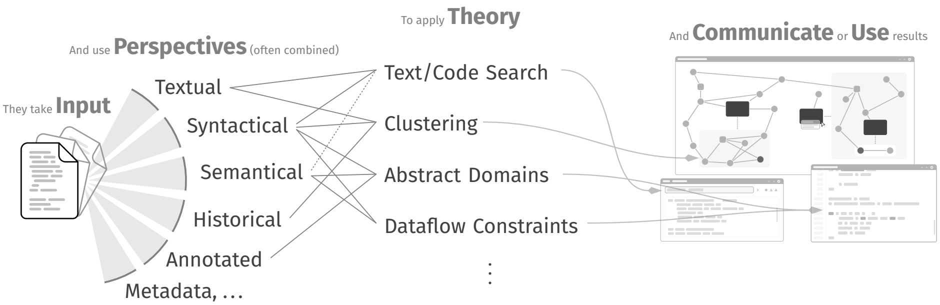

Static Analysis in the Real World

Software Quality Assurance — Static Code Analysis, III | Florian Sihler | December 17, 2025

Software Engineering
Programming Languages

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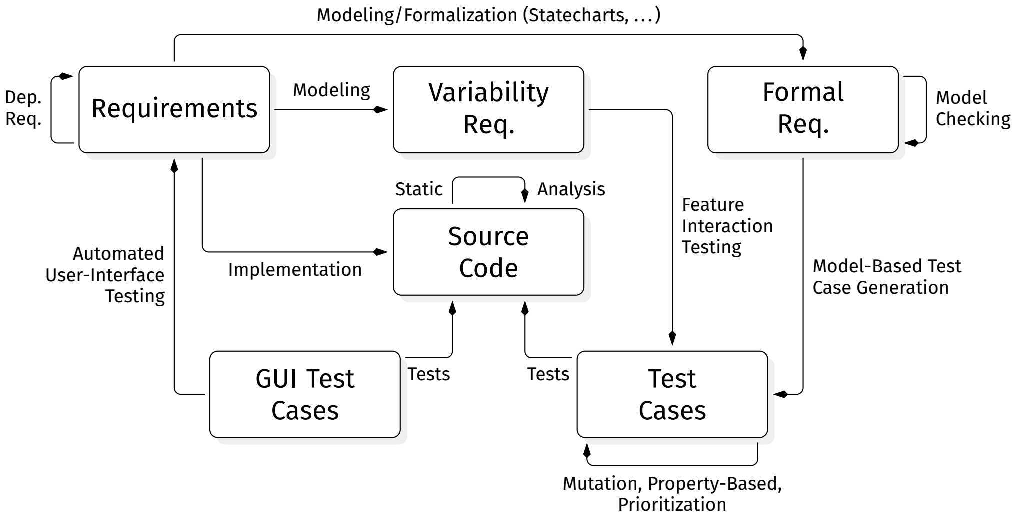
1. A Small Recap

Embedding a Landscape



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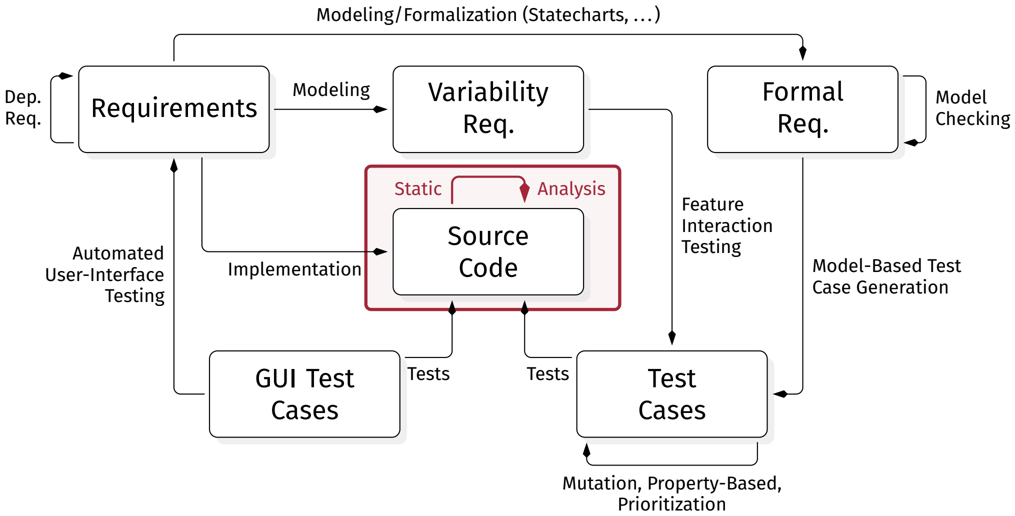

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A way to soundly over-approximate all possible program behaviors.

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   "We add the lower bounds and the upper bounds, assuming the environment ρ."
   (assuming a language like Java and ignoring overflows)

Recap: Applying the Interval Domain

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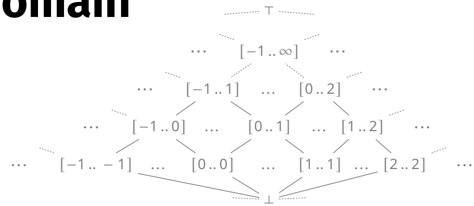

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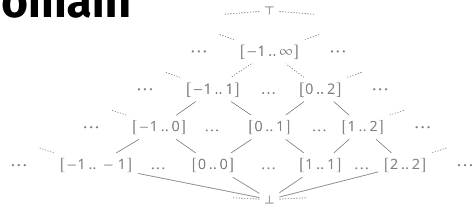

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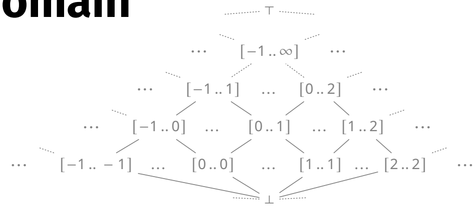

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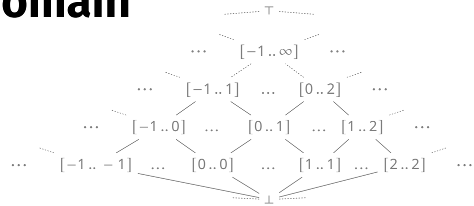

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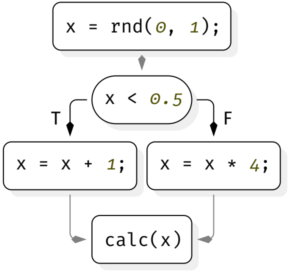

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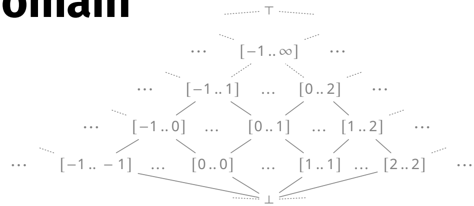

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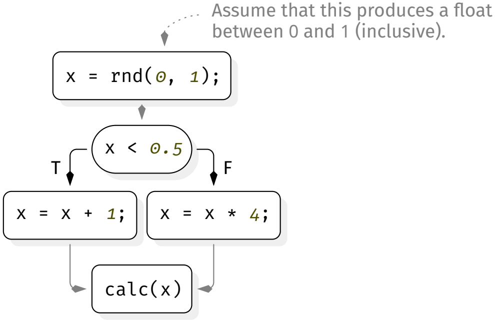

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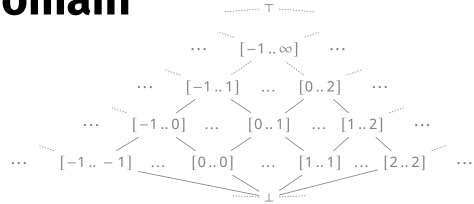

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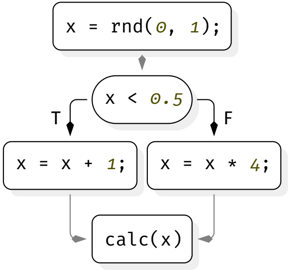

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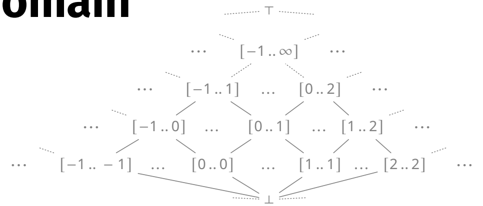

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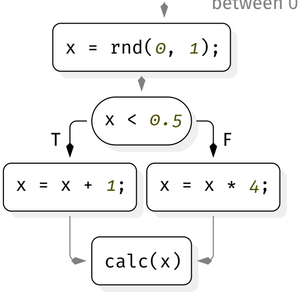

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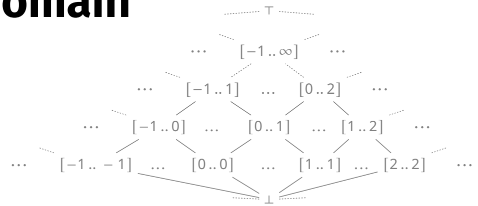

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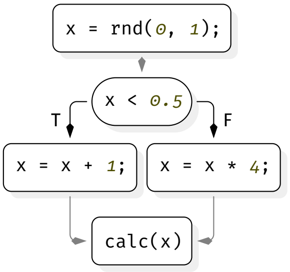

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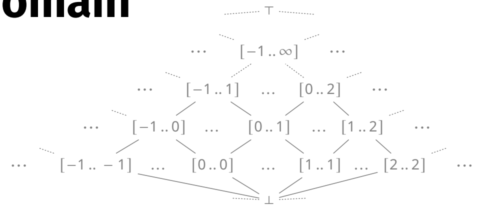

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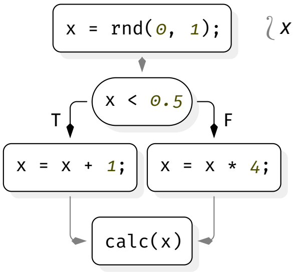

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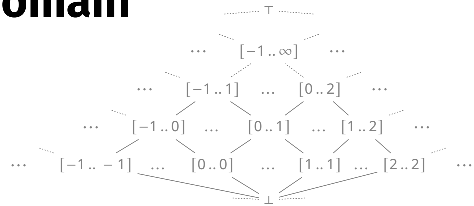

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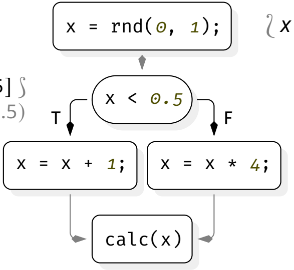

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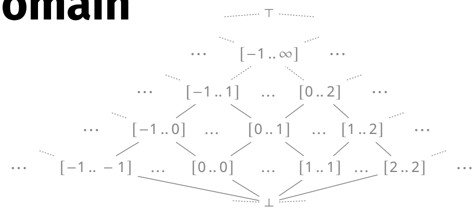

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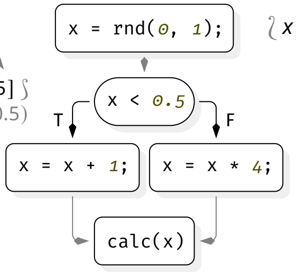

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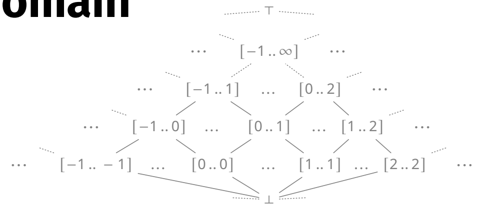

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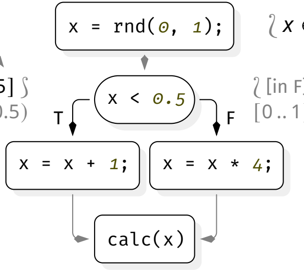

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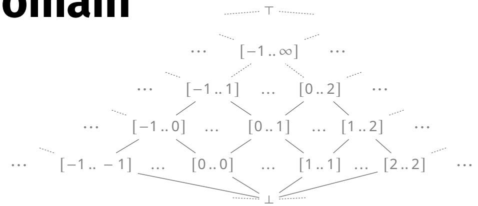

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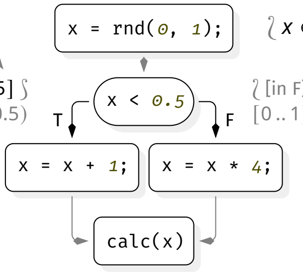

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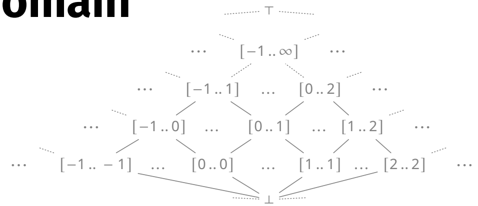
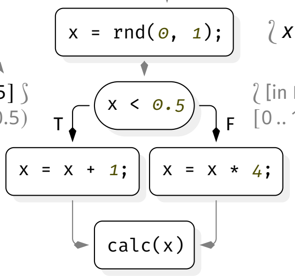
Recap: Applying the Interval Domain



6. Do we need widening (∇) for this domain?
   Yes, because the lattice contains infinite ascending chains, e.g., $[0 \,..\, 0] \sqsubseteq [0 \,..\, 1] \sqsubseteq [0 \,..\, 2] \sqsubseteq \dots$

Assume that this produces a float between 0 and 1 (inclusive).

Over-Approx. to ignore the 0.499… problem



$\wr\, x \in [0 \,..\, 1] \,\wr$

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$[0 \,..\, 1] \cap [0 \,..\, 0.5)$

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$\wr\, x \in [1 \,..\, 1.5] \,\wr$

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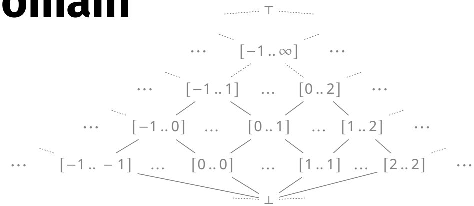

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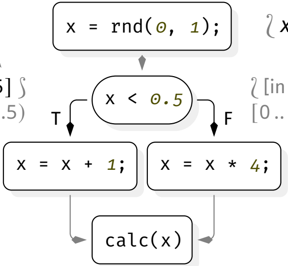

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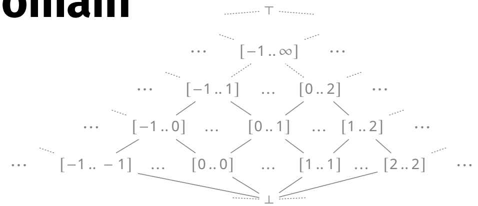
Recap: Applying the Interval Domain



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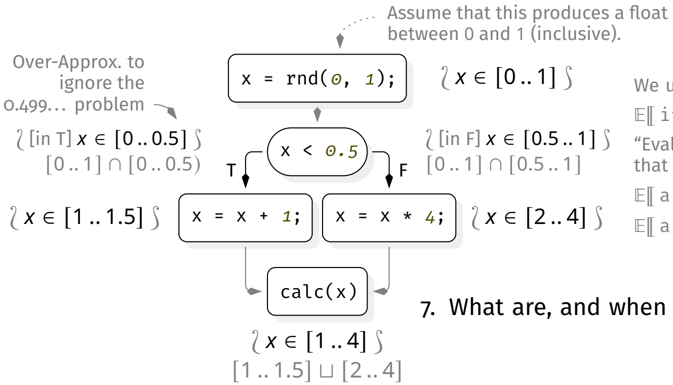

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Recap: Applying the Interval Domain

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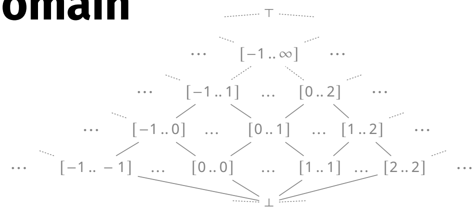

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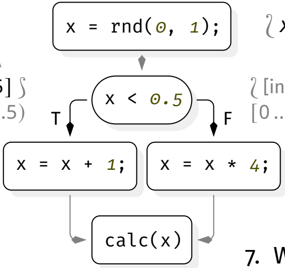

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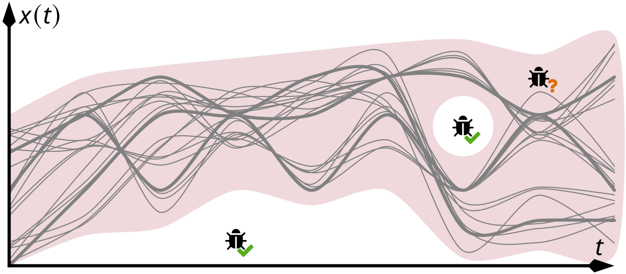
7. What are, and when do we need fixpoints?
   $f(x) = x$. *We reach fixpoints when the repeated application of semantics produces no new info. When analyzing loops or recursive functions, we compute fixpoints to find a stable state (of all code executions).*

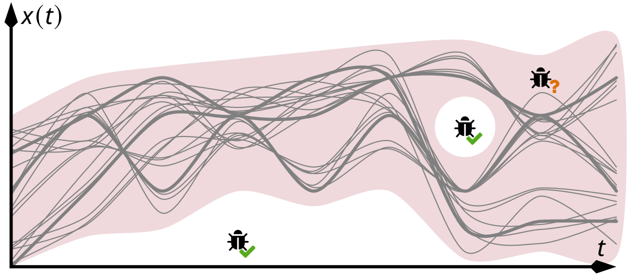
2. Introduction

What we have…

What we have...

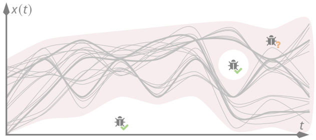


What we have...



Abstractions

What we have...



Abstractions

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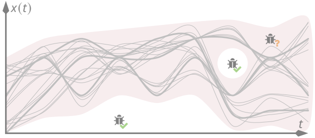

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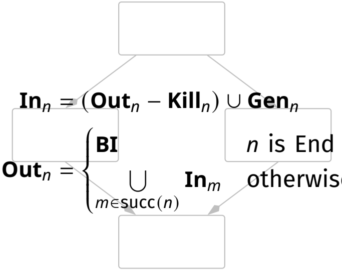

$$\mathbf{In}_n = (\mathbf{Out}_n - \mathbf{Kill}_n) \cup \mathbf{Gen}_n$$

$$\mathbf{Out}_n = \begin{cases} \mathbf{BI} & n \text{ is End} \\ \bigcup\limits_{m \in \mathrm{succ}(n)} \mathbf{In}_m & \text{otherwise} \end{cases}$$

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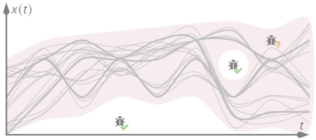

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Data Flow Analysis

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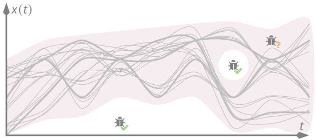
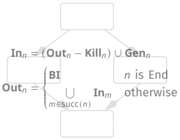


Abstractions



Data Flow Analysis

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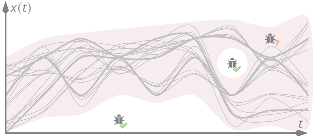

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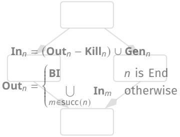
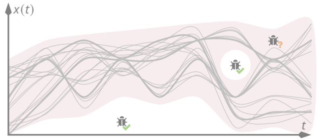


Data Flow Analysis

…

What we have… **Theory**



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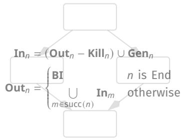


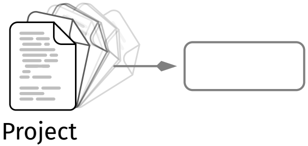
Data Flow Analysis

…

What we want...

What we want… **Tools**

What we want… Tools



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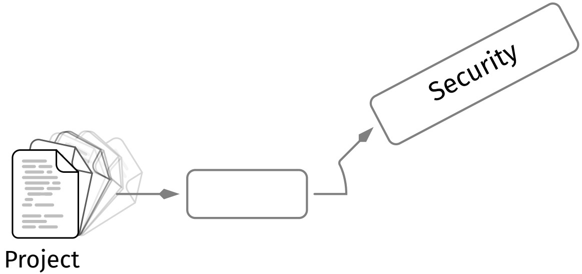

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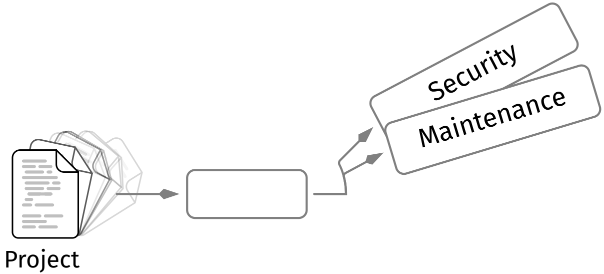

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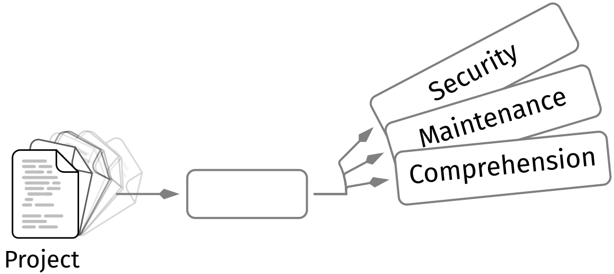

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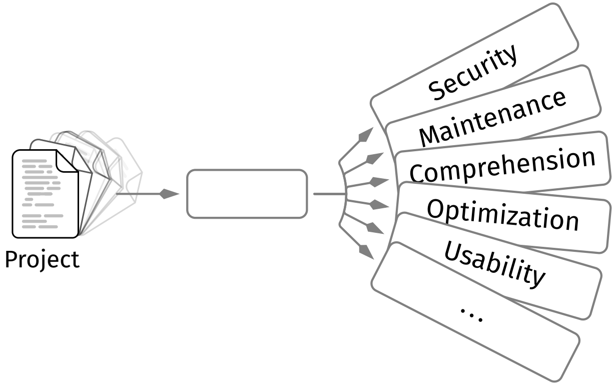

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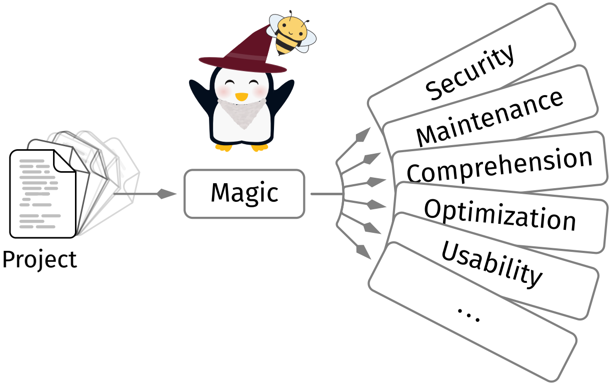

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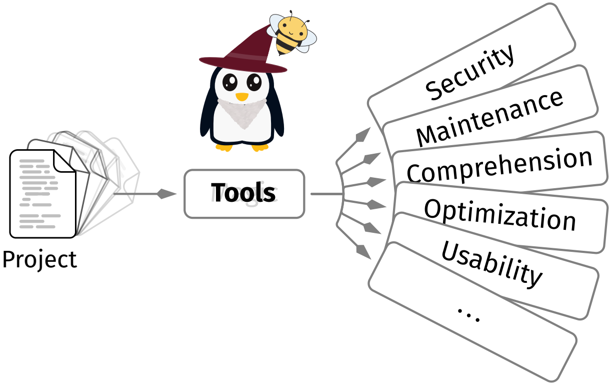

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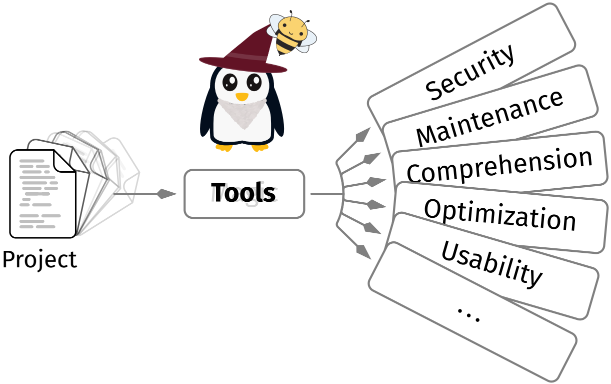

"Any sufficiently advanced technology is indistinguishable from magic." — Arthur C. Clarke

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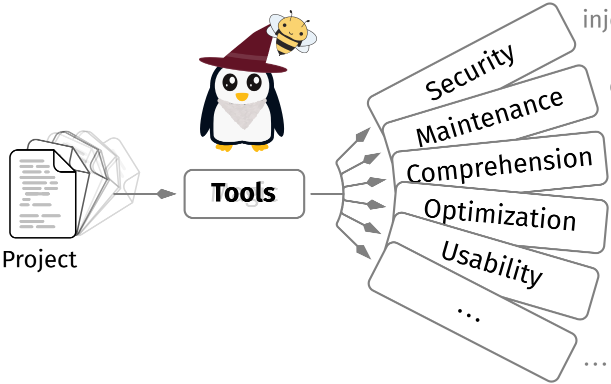

injections, leaks, …

code clones, deprecation, …

documentation, structure, …

better algorithms, memoization, …

color deficiency, size, …

…

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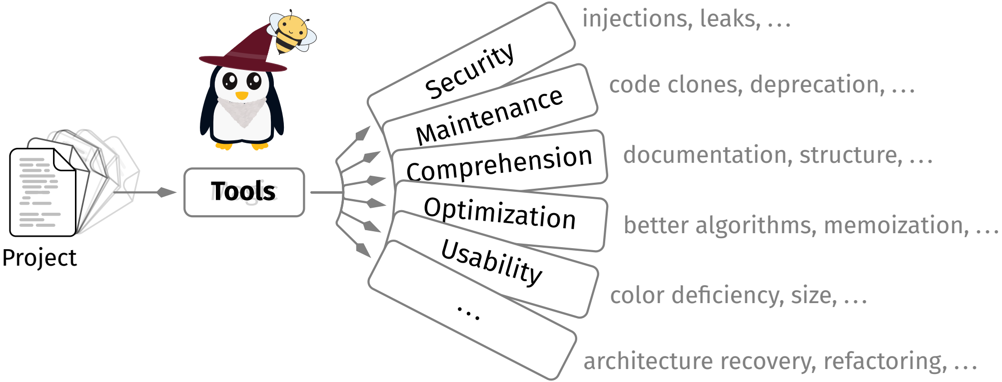
What we want… **Tools**



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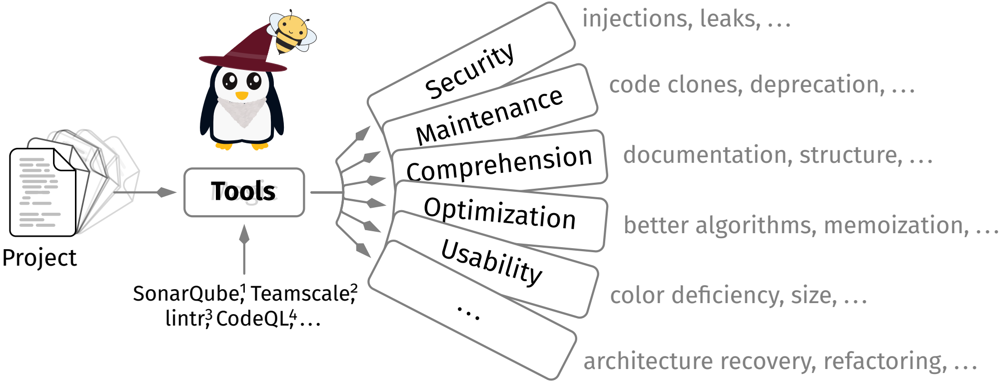

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[1] sonarsource.com, [2] teamscale.com, [3] lintr.r-lib.org, [4] codeql.github.com

What do they… do?

What do they… do?

They take **Input**

What do they… do?

And use **Perspectives** (often combined)

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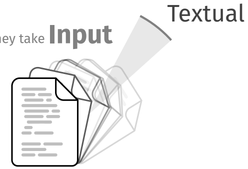

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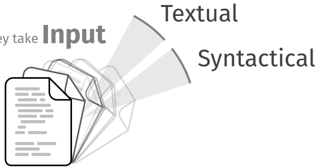

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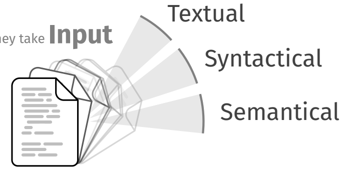

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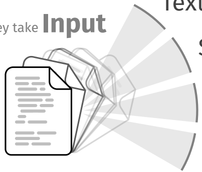

Textual

Syntactical

Semantical

Historical

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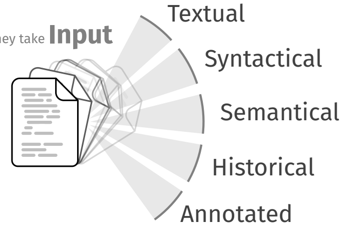

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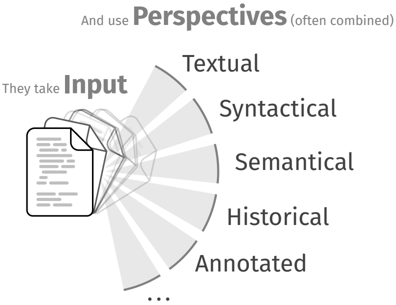

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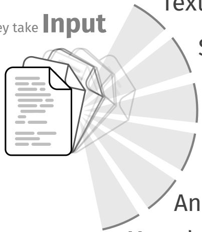

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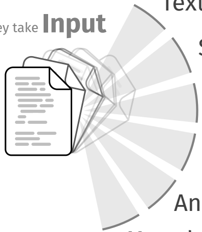

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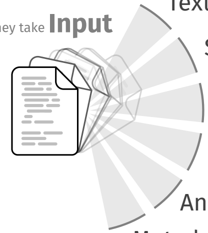

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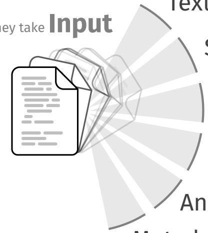
To apply **Theory**

What do they… do?

And use **Perspectives** (often combined)

To apply **Theory**

They take **Input**



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Text/Code Search

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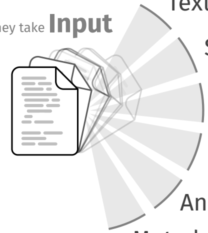
And use **Perspectives** (often combined)

They take **Input**



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To apply **Theory**

Text/Code Search

Clustering

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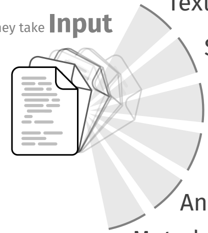

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To apply **Theory**

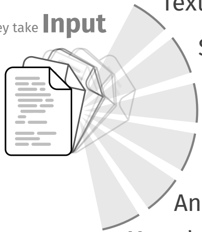
Text/Code Search

Clustering

Abstract Domains

What do they… do?

They take **Input**



And use **Perspectives** (often combined)

Textual

Syntactical

Semantical

Historical

Annotated

Metadata, …

Some of those are the
result of other static
or dynamic analyses

To apply **Theory**

Text/Code Search

Clustering

Abstract Domains

Dataflow Constraints

⋮

What do they… do?

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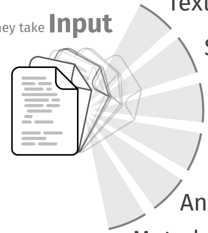

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- ⋮

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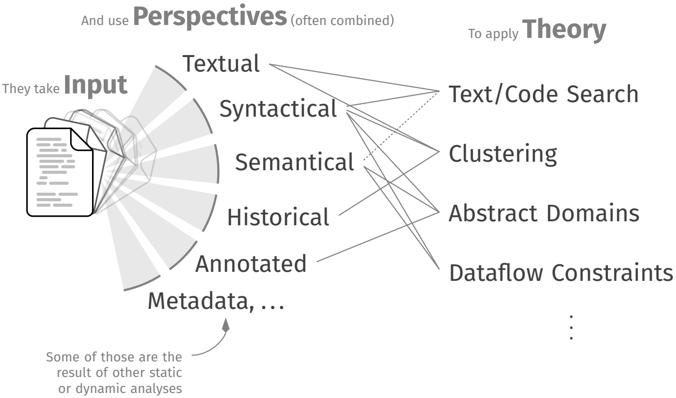

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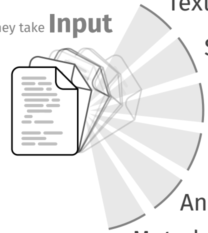

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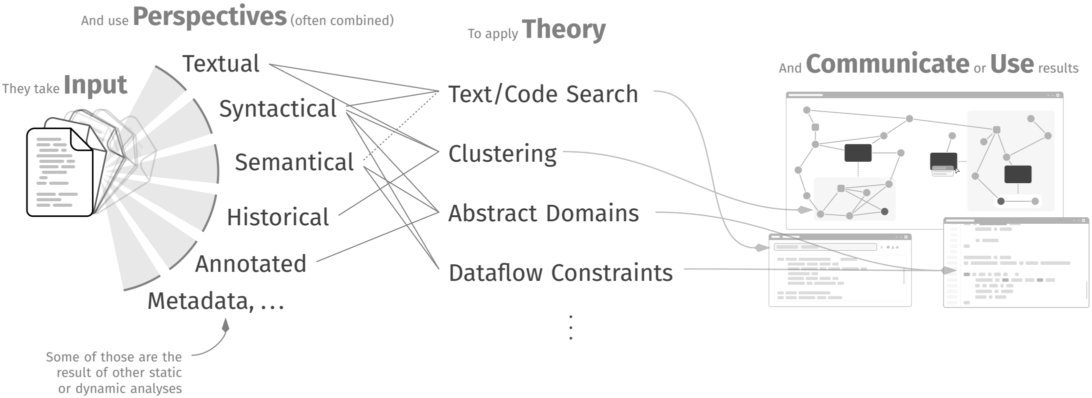
And **Communicate** or **Use** results

What do they… do?



3. Real-World Static Analyzers

Let’s Look at Tools

Comprehension

Security

Optimization

Usability

Maintenance

Let's Look at Tools

github.com/analysis-tools-dev/static-analysis

Let’s Look at Tools

sonar

github.com/analysis-tools-dev/static-analysis

Let's Look at Tools

Teamscale

sonar

github.com/analysis-tools-dev/static-analysis

Let’s Look at Tools

SootUp

flowR

LiquidHaskell

AbsInt

lintr

Teamscale

Infer

LiSA

CodeQL

sonar

ESLint

MOPSA analyzer

frama C

github.com/analysis-tools-dev/static-analysis

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There are countless…

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- SonarLint

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- Astrée

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- Astrée (sound!)

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- Java Language Server

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github.com/SonarSource/sonar-java

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  - Widely varying support and rules

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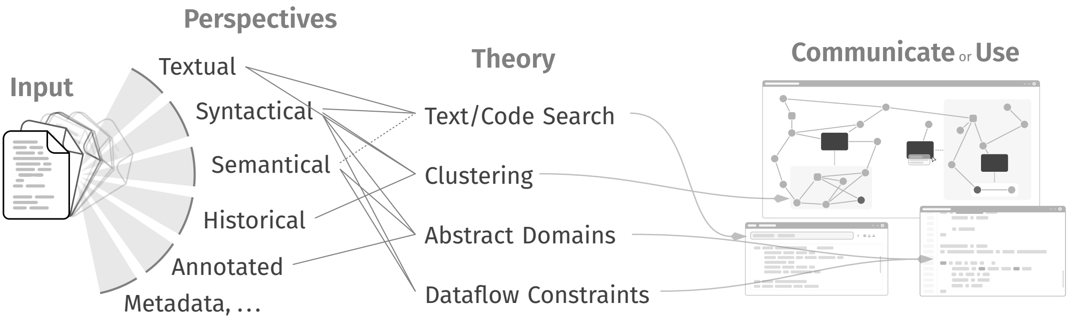

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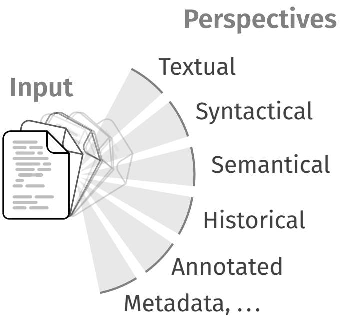

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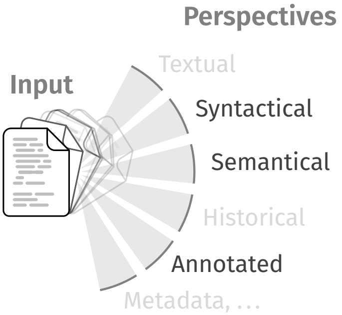

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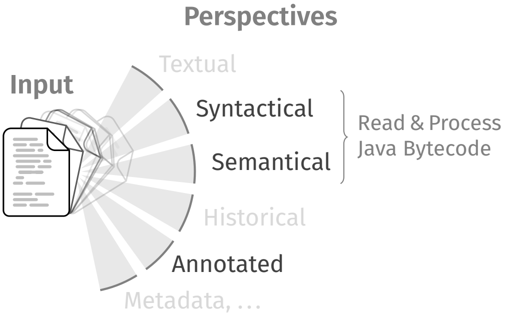

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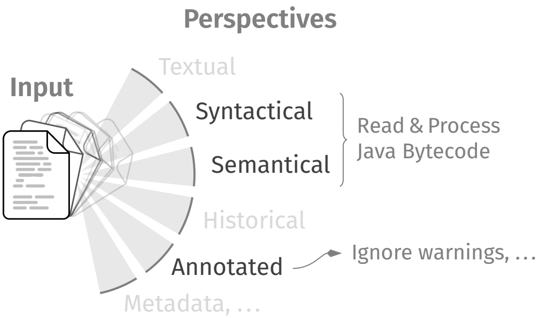

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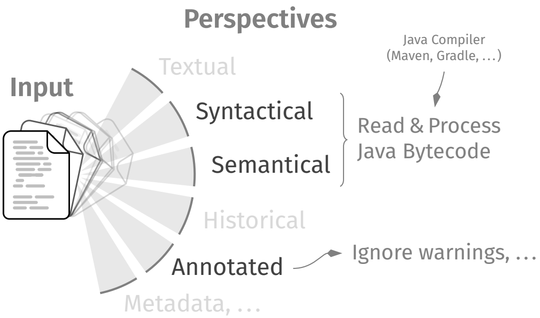

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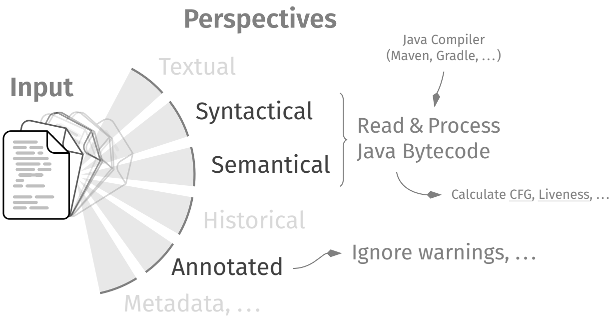

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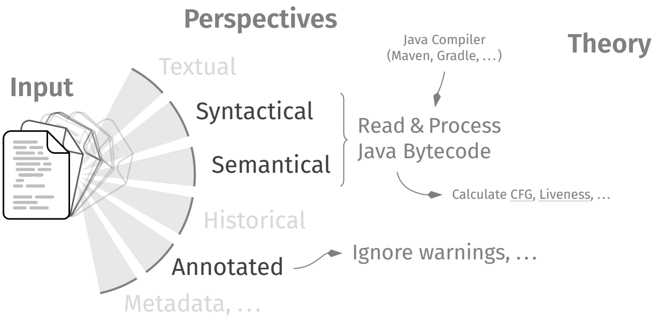

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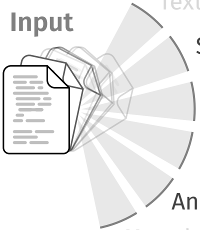

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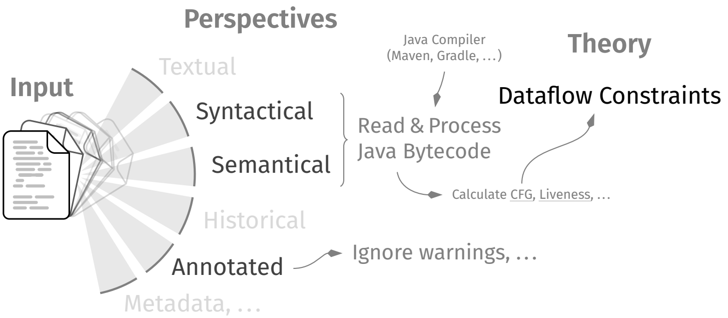

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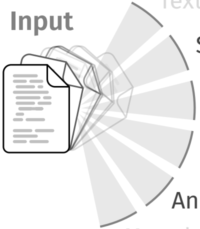
Perspectives

Input



Textual

Syntactical

Semantical

Historical

Annotated

Metadata, ...

Java Compiler
(Maven, Gradle, ...)

Read & Process
Java Bytecode

Calculate CFG, Liveness, ...

Theory

Dataflow Constraints

Ignore warnings, ...

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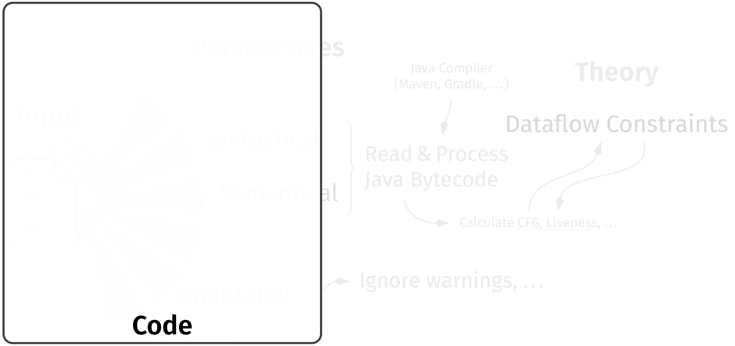

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Control Flow Graph

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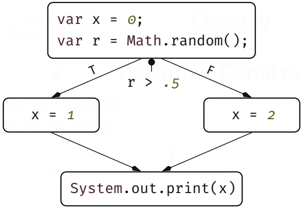

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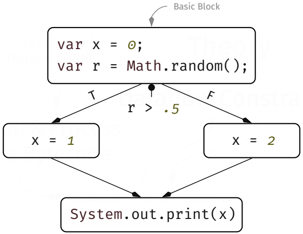

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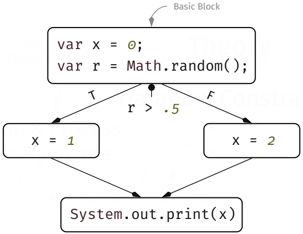

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With fixpoint iteration

Liveness[KSK09, p. 26]

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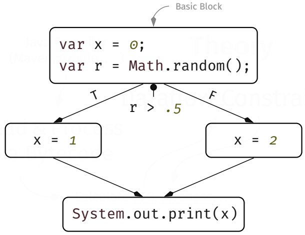

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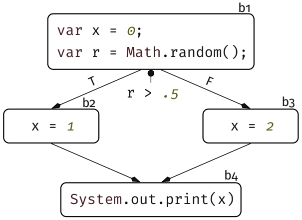

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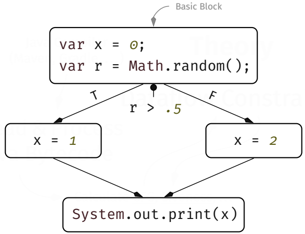

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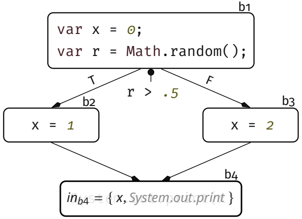

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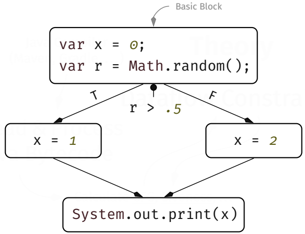

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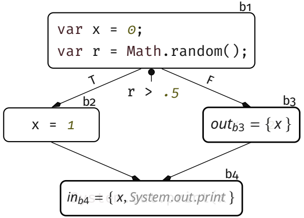

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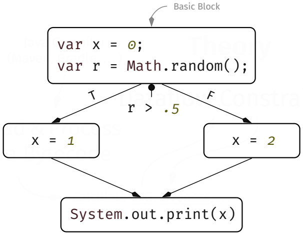

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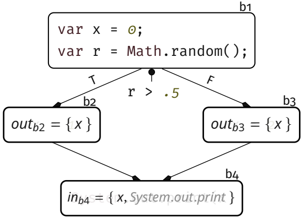

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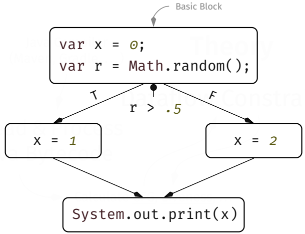

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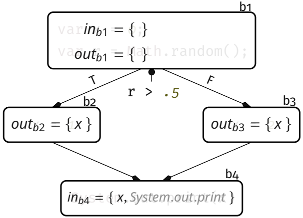

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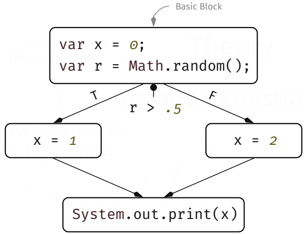
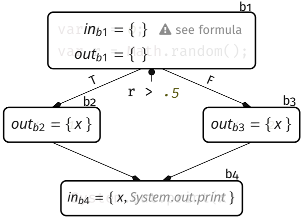


Control Flow Graph



Liveness[KSK09, p. 26]

github.com/SonarSource/sonar-java

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SonarLint

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  - Widely varying support and rules
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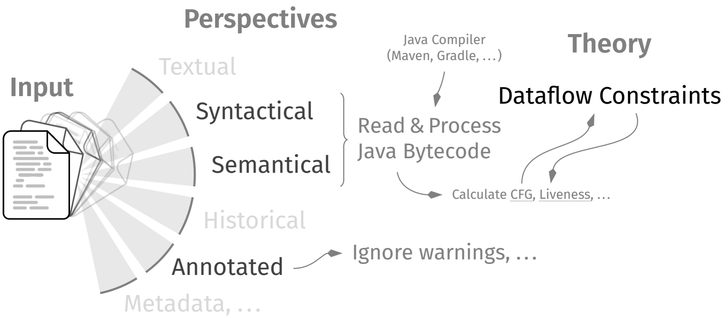
Separate frontends and analyzers per language!

733 rules!



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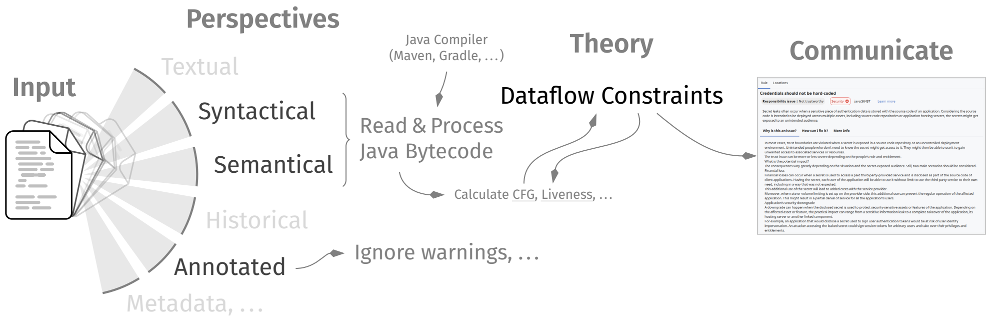

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SonarLint — **Unused Assignments**

Implementation of Rule S1854

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```
LiveVariables liveVariables = LiveVariables.analyze(cfg);
// Liveness analysis provides information only for block boundaries,
    so we should do analysis between elements within blocks
for(CFG.Block block : cfg.blocks()) {
  checkElements(block, liveVariables.getOut(block), methodSymbol);
}
```

Implementation of Rule S1854

SonarLint — Hardcoded Credentials

Implementation of Rule S6437

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  visiting the dataflow links and checking for predefined "plain text"

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- Check if the arguments are “constant”
  visiting the

```java
for (int targetArgumentIndex : method.indices) {
  ExpressionTree argument = arguments.get(targetArgumentIndex);
   var secondaryLocations = new ArrayList<JavaFileScannerContext.Location>();
   if (isExpressionDerivedFromPlainText(argument,
             secondaryLocations, new HashSet<>())) {
      reportIssue(argument, ISSUE_MESSAGE, secondaryLocations, null);
   }
}
```

Implementation of Rule S6437

Let’s Look at Tools

SootUp

flowR

LiquidHaskell

AbsInt

lintr

Teamscale

Infer

LiSA

CodeQL

sonar

Clang

GCC

ESLint

MOPSA analyzer

frama C

There are countless…

github.com/analysis-tools-dev/static-analysis

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Astrée

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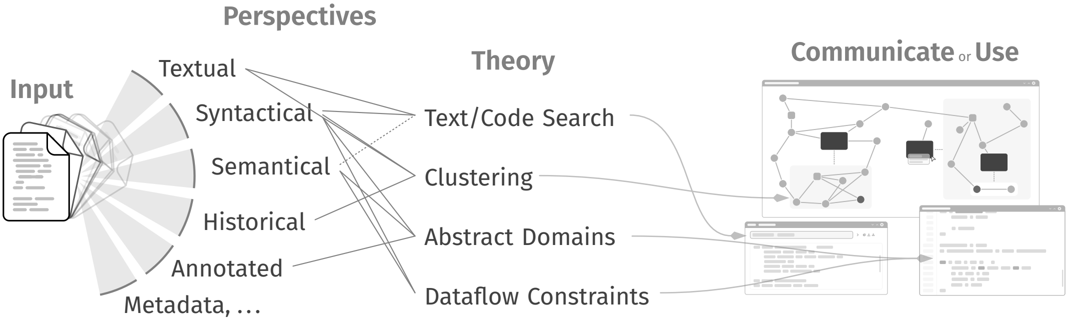

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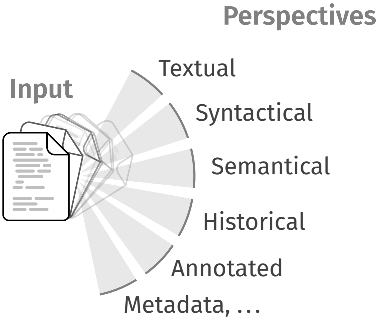

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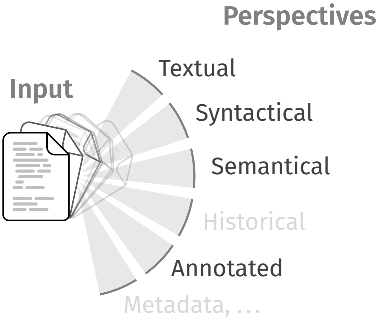

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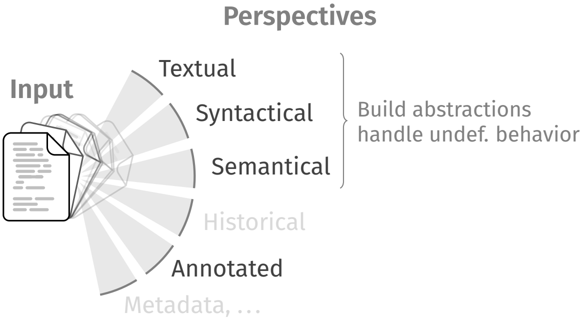

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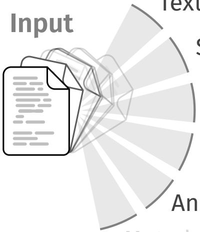

Build abstractions
handle undef. behavior

Specifications, Configuration, …
for widening, assumptions, …

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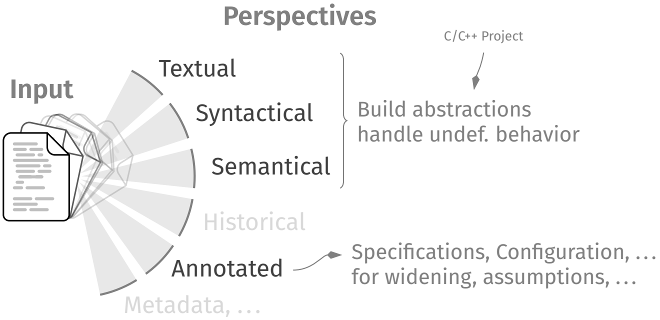

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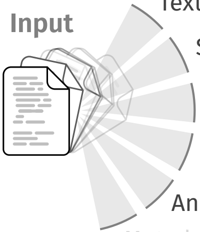

Textual
Syntactical
Semantical

C/C++ Project

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CFG, Data Races, …

Historical

Annotated

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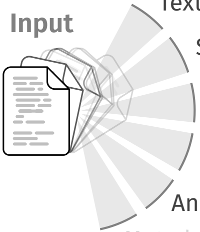

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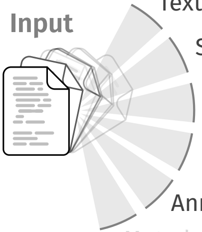

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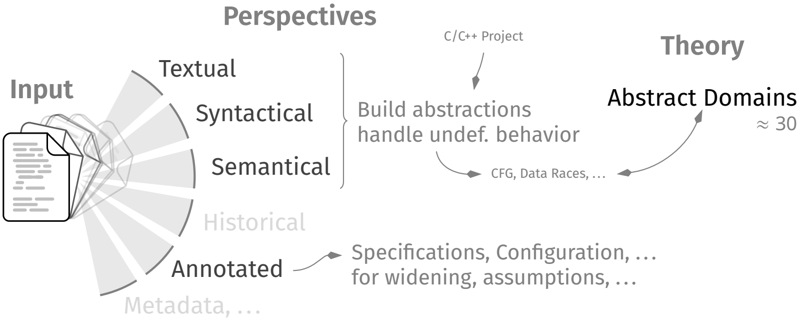

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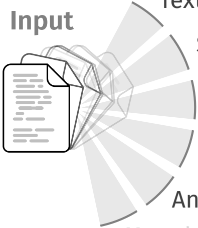

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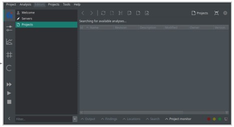

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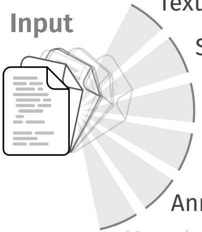

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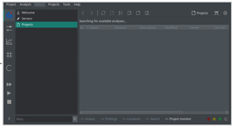

Analysis Time
from couple of hours to days

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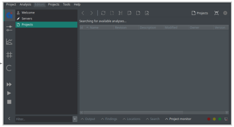

Local precision tuning

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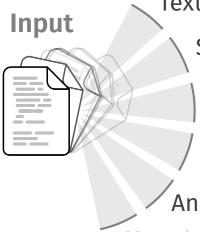

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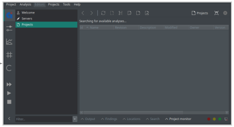

Semantic Loop Unrolling

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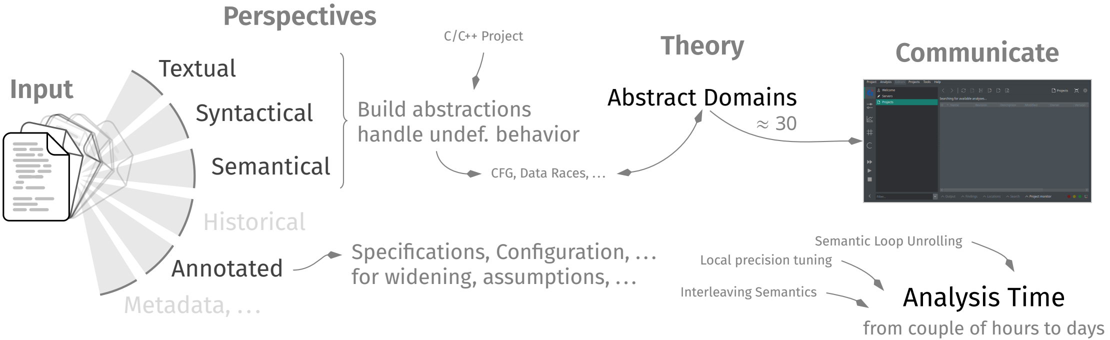

Let’s Look at Tools

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flowR
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AbsInt
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LiSA
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GCC
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There are countless…

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LiSA

github.com/lisa-analyzer/lisa

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Similar to Mopsa or Apron

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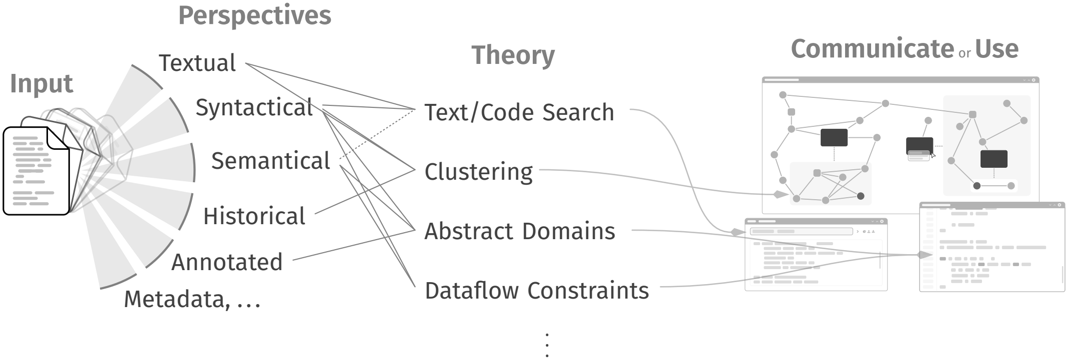

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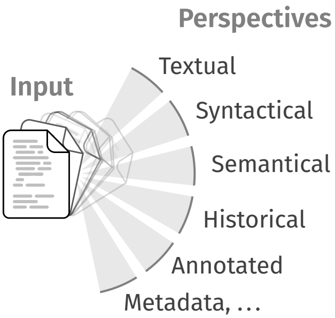

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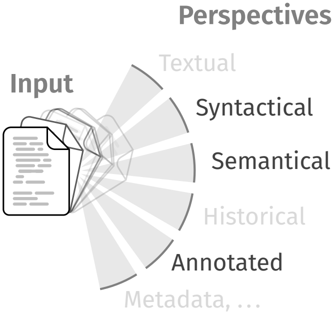

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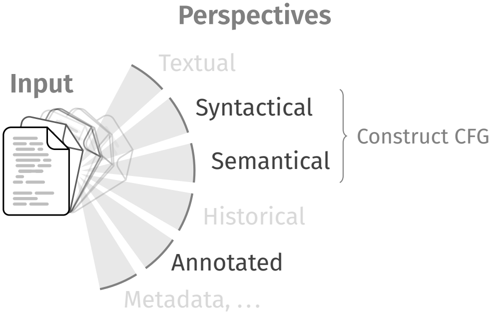

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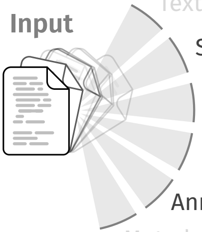

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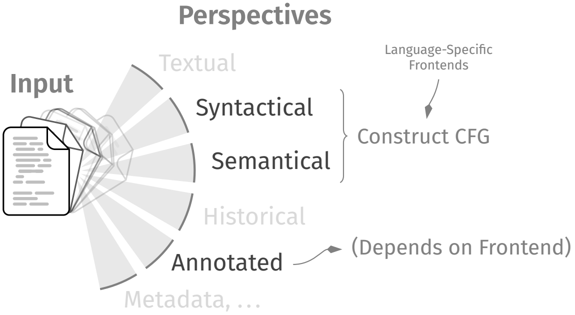
- Custom frontends for Rust, Go, Python, EVM, ...



github.com/lisa-analyzer/lisa

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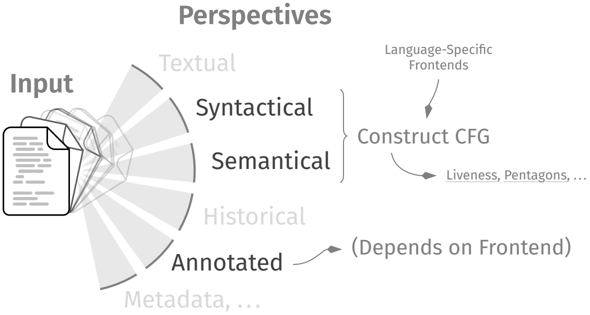

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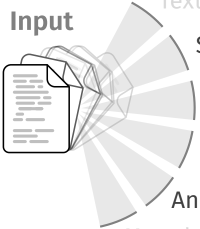
Perspectives

Input



Textual

Syntactical

Semantical

Construct CFG

Language-Specific Frontends

Liveness, Pentagons, ...

Historical

Annotated — (Depends on Frontend)

Metadata, ...

Theory

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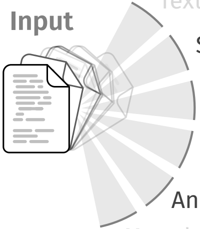

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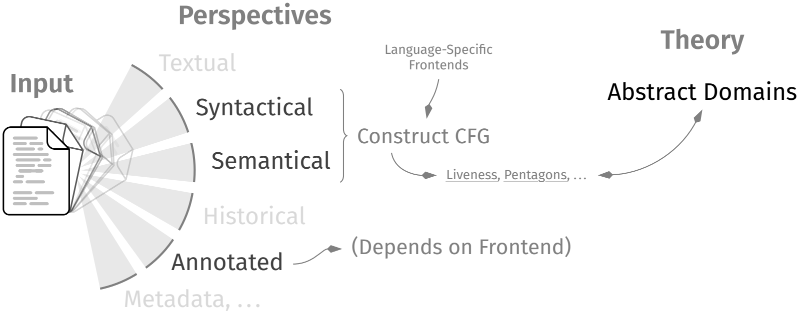

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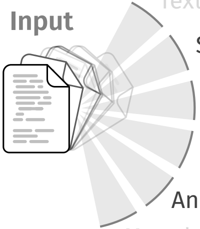

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Theory

Abstract Domains

Use

Extension Interfaces
Fixpoint Iterations,
…

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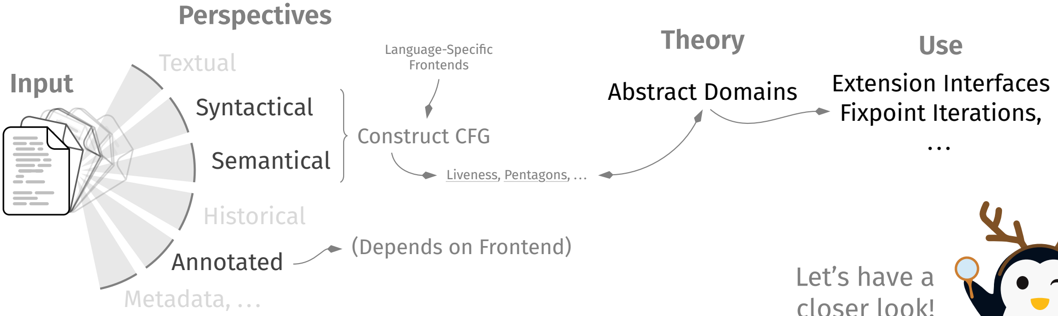

Let's have a closer look!

github.com/lisa-analyzer/lisa

LiSA — Interval Analysis[Cou21, p. 389]

[Cou21] “Principles of Abstract Interpretation” (Cousot)

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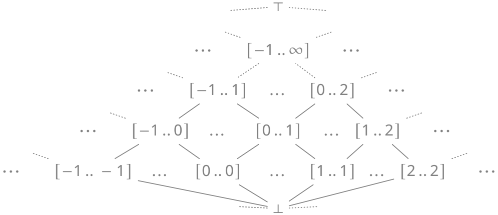

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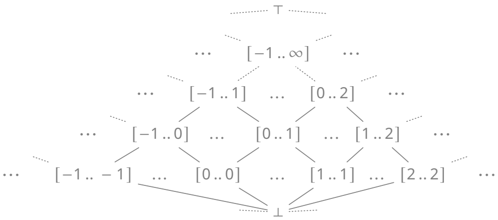

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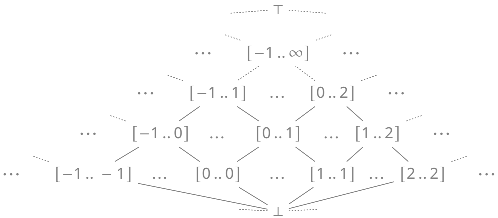

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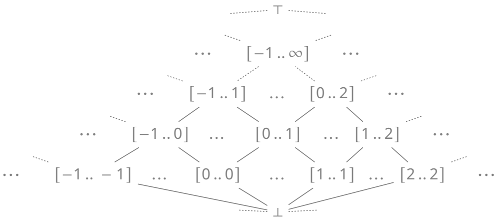

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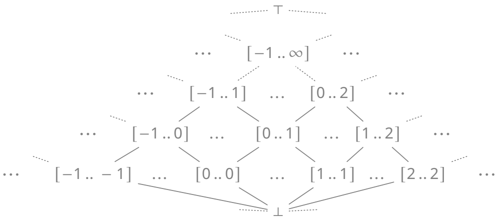

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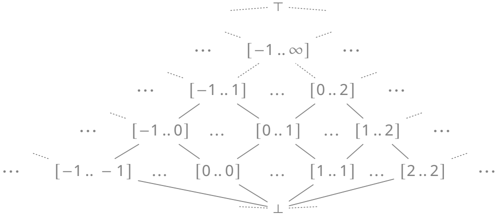

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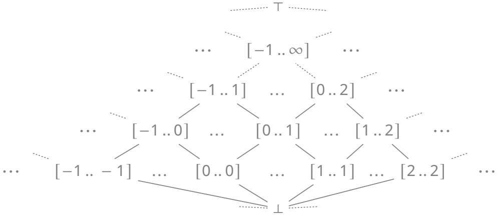

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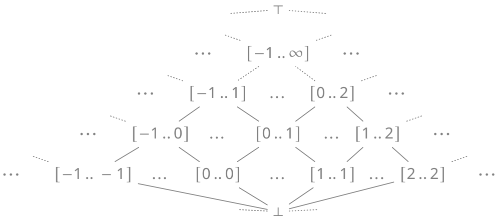

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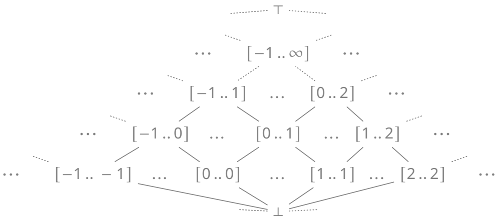

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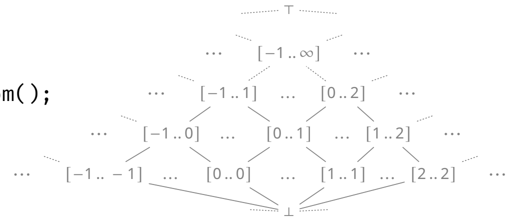

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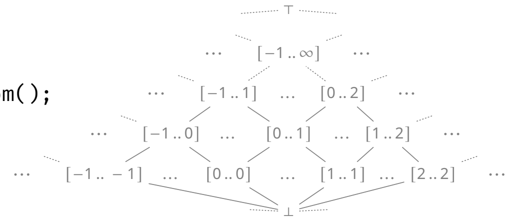

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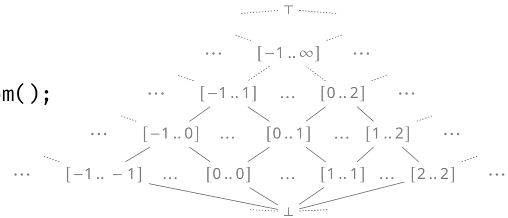

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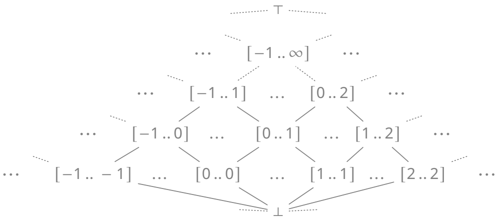
LiSA — Interval Analysis[Cou21, p. 389] Semantics



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LiSA — Interval Analysis[Cou21, p. 389] Semantics

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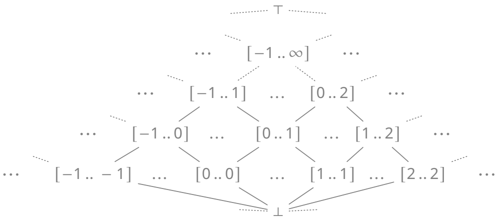

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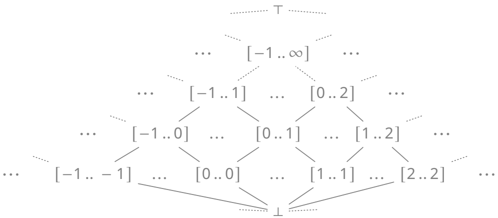

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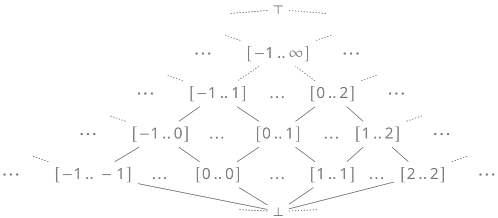

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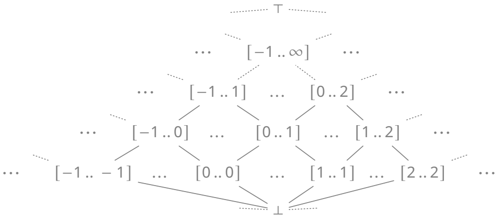

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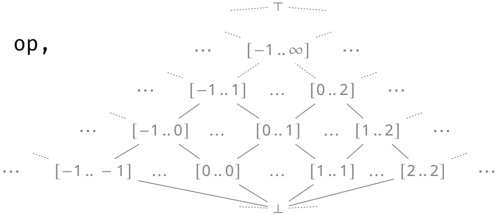

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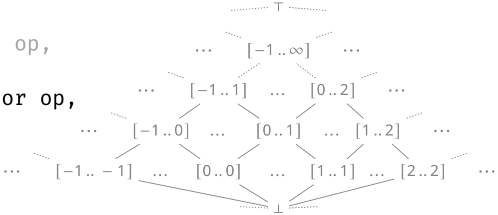

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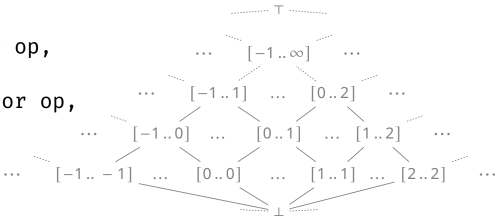

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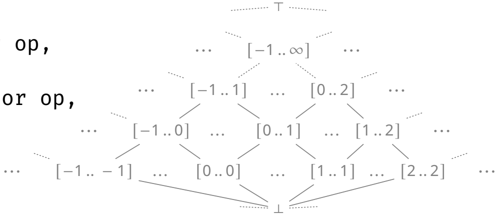
Fold-like Evaluation



[Cou21] “Principles of Abstract Interpretation” (Cousot)

Let's Look at Tools

SootUp

flowR

LiquidHaskell

AbsInt

lintr

Teamscale

Infer

LiSA

CodeQL

sonar

Clang

GCC

ESLint

MOPSA analyzer

frama C

There are countless…

github.com/analysis-tools-dev/static-analysis

Let's Look at Tools

- SonarLint
- Astrée (sound!)
- LiSA

- **Java Language Server**
- lintr
- flowR

There are countless…

github.com/analysis-tools-dev/static-analysis

Java Language Server

github.com/eclipse-jdtls/eclipse.jdt.ls

Java Language Server

- Uses the Language Server Protocol to provide static analysis for Java

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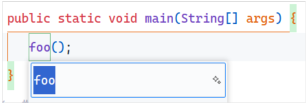


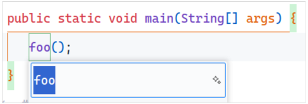
Rename Refactoring

github.com/eclipse-jdtls/eclipse.jdt.ls

Java Language Server

- Uses the Language Server Protocol to provide static analysis for Java

Code Actions



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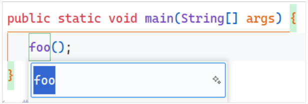

Rename Refactoring

...

github.com/eclipse-jdtls/eclipse.jdt.ls

Java Language Server

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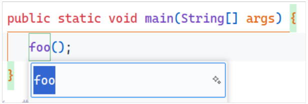
Code Actions



Rename Refactoring

...

- Relies on the Eclipse JDT Language Server

github.com/eclipse-jdtls/eclipse.jdt.ls

Java Language Server

- Uses the Language Server Protocol to provide static analysis for Java
  Renaming, Code Actions, ...
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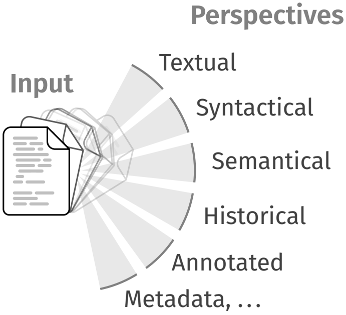

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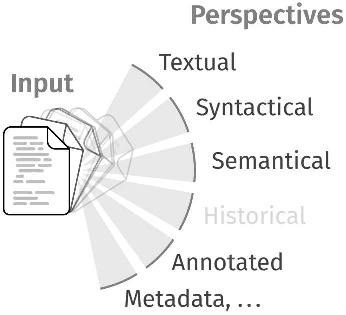

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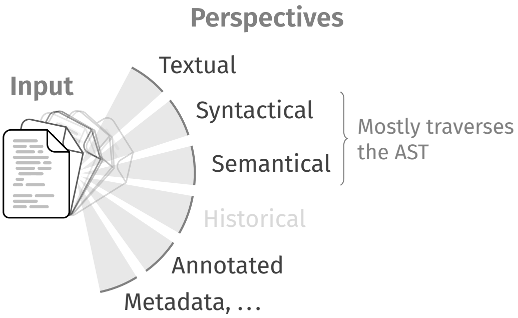

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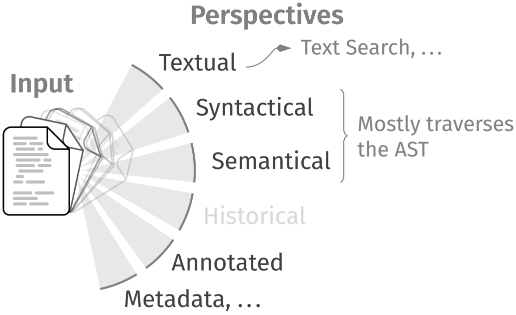

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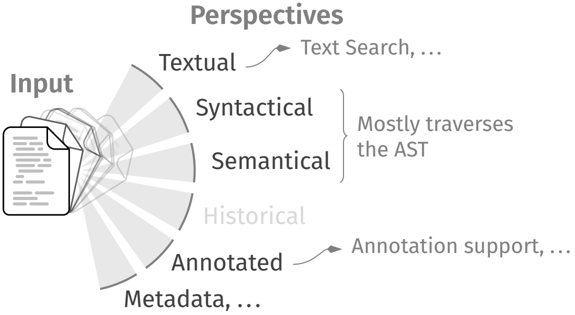

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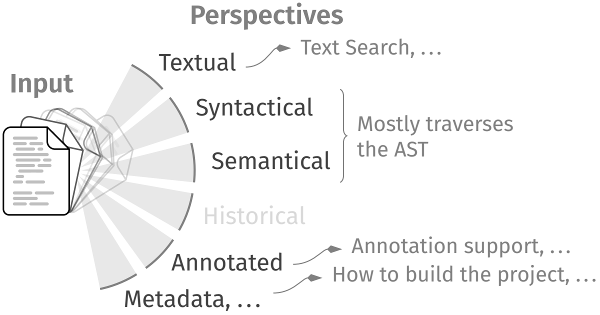

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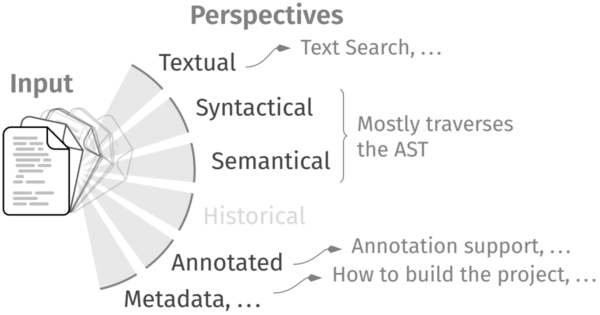

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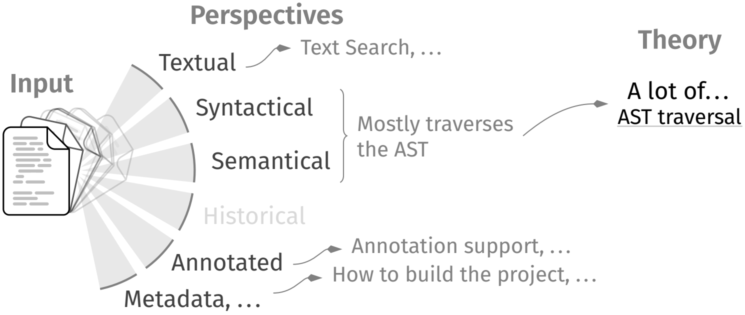

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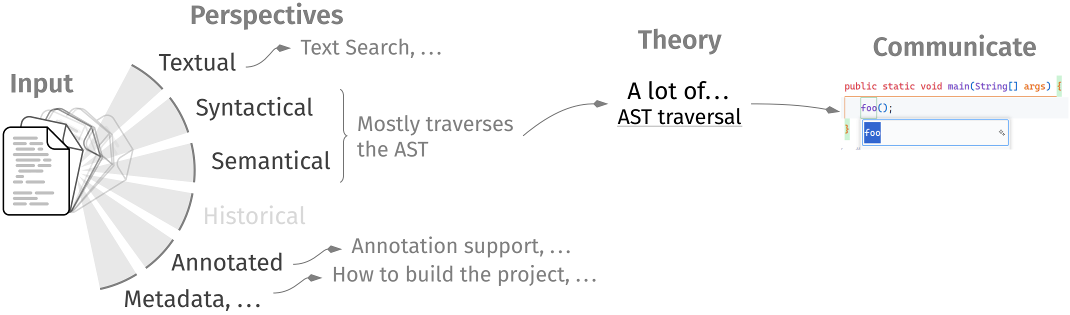

github.com/eclipse-jdtls/eclipse.jdt.ls

Let’s Look at Tools

SootUp

flowR

LiquidHaskell

AbsInt

lintr

Teamscale

Infer

LiSA

CodeQL

sonar

Clang

GCC

ESLint

MOPSA analyzer

frama C

There are countless…

github.com/analysis-tools-dev/static-analysis

Let's Look at Tools

- SonarLint
- Astrée (sound!)
- LiSA
- Java Language Server
- **lintr**
- flowR

There are countless…

github.com/analysis-tools-dev/static-analysis

lintr

github.com/r-lib/lintr

[Flü+19] “R melts brains: an IR for first-class environments and lazy effectful arguments” (Flückiger et al.)

lintr

- A linter for the R programming language

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x <- 4
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f() # 4
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```r
x <- 4
f <- function() x
body(f) <- quote(y)
y <- 42
f()
```

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```

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```
'if' <- function(...) 42
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`if` <- function(...) 42
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- Common static analysis strategies have their… problems with R

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  - ~~Control Flow Graphs?~~
  - AST ~~Traversal?~~ (mostly) Pattern Matching and Evaluation!

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lintr — Under the Hood

≈ 100 rules

lintr — Under the Hood

≈ 100 rules

```
magic <- "
   expr[LEFT_ASSIGN or EQ_ASSIGN]/expr[2][FUNCTION or OP-LAMBDA]
   | expr_or_assign_or_help[EQ_ASSIGN]/expr[2][FUNCTION or OP-LAMBDA]
   | equal_assign[EQ_ASSIGN]/expr[2][FUNCTION or OP-LAMBDA]
   | //SYMBOL_FUNCTION_CALL[text() = 'assign']/parent::expr/following-sibling::
       expr[2][FUNCTION or OP-LAMBDA]
   | //SYMBOL_FUNCTION_CALL[text() = 'setMethod']/parent::expr/following-sibling::
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- What does this do?

lintr — Under the Hood

≈ 100 rules

R/object_usage_linter.R

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- What does this do?
  This XPATH expression matches assignments to functions

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≈ 100 rules

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- What does this do?
  This XPATH expression matches assignments to functions
- It is very rigid (no alias tracking, flow sensitivity, …)

lintr — Under the Hood

≈ 100 rules

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- What does this do?
  This XPATH expression matches assignments to functions
- It is very rigid (no alias tracking, flow sensitivity, ...)
- And it...

lintr — Under the Hood

≈ 100 rules

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lintr — Under the Hood

≈ 100 rules

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- What does this do?
  This XPATH expression matches assignments to functions
- It is very rigid (no alias tracking, flow sensitivity, …)
- And it… cheats:
  `try_silently(eval(envir = env,parse(text = code, keep.source = TRUE)))`

lintr — Under the Hood

≈ 100 rules

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```
magic <- "
   expr[LEFT_ASSIGN or EQ_ASSIGN]/expr[2][FUNCTION or OP-LAMBDA]
   | expr_or_assign_or_help[EQ_ASSIGN]/expr[2][FUNCTION or OP-LAMBDA]
   | equal_assign[EQ_ASSIGN]/expr[2][FUNCTION or OP-LAMBDA]
   | //SYMBOL_FUNCTION_CALL[text() = 'assign']/parent::expr/following-sibling::
       expr[2][FUNCTION or OP-LAMBDA]
   | //SYMBOL_FUNCTION_CALL[text() = 'setMethod']/parent::expr/following-sibling::
       expr[3][FUNCTION or OP-LAMBDA]"
```

- What does this do?
  This XPATH expression matches assignments to functions
- It is very rigid (no alias tracking, flow sensitivity, …)
- And it… cheats:
  `try_silently(eval(envir = env,parse(text = code, keep.source = TRUE)))`
  It simply runs (parts of) the program (including side-effects), …

Let's Look at Tools

SootUp

flowR

LiquidHaskell

AbsInt

lintr

Teamscale

Infer

LiSA

CodeQL

sonar

Clang

GCC

ESLint

MOPSA analyzer

frama C

There are countless…

github.com/analysis-tools-dev/static-analysis

Let's Look at Tools

- SonarLint
- Astrée (sound!)
- LiSA
- Java Language Server
- lintr
- **flowR**

There are countless…

github.com/analysis-tools-dev/static-analysis

Existing Work

Existing Work

		references	op. assignments	dyn. assignments	value trace	control flow	non-std. eval.	special operators	function calls	libraries	reflection	side effects	static scope	dynamic scope	type inference	pointer analysis	external files	pre-processors	hooks	FFI	expose info.	goal	method
	flowR	●	●	○	○	●	○	●	●	○	○	○	●	○			○				●	static analysis	AST visitor
[Cha23]	{checkglobals}	●	●	○			○	○	○	○			○							○		missing libs.	AST visitor
[Lan+23]	{CodeAnalysis}	●	●	○	○	●	○	○	○	○			●		○		○				●	static analysis	AST visitor
[Lan+18]	{CodeDepends}	●	●	○			○	○	○	○		○	●								●	static analysis	AST visitor
[Tie23]	{codetools}	●	●	○	○	○	○	○	○				●								●	static analysis	AST visitor
[Csa23]	{cyclocomp}					○							●									static analysis	AST visitor
[Rob16]	{DependenciesGraphs}								○	○											○	visualize	regex
[Kar21]	{dfgraph}	●	○			○			○												○	static analysis	AST visitor
[Fab23]	{flow}	●	○			●		○		○							○				○	vis., debug	AST visitor
[App24]	{foodwebr}								○												○	visualize	regex
[Ben22]	{globals}	●	●		○	○	○	○	○				●									distributed env.	AST visitor
[Lai23]	{languageserver}	●	●	○					○	○			●								○	editor support	XPath, visitor
[Hes+23]	{lintr}	●	●	○		○		○		○			●									linting	XPath, visitor
[Bra18]	{mvbutils}								○													utility	regex
[van23]	{PaRe}		○			○			○													code review	regex
[Pad21]	{pkgstats}	●	○					○	○				○									pkg. insights	ctags & gtags
[Lau22]	{Rclean}	●	●	○			○	○	○	○			●									debug/refactor	PDG traversal
[Rod21]	{rco}	●	●		●	○							●									optimization	AST visitor
[Pat19]	{rflowgraph}	●							○	○											○	call graph	AST visitor
[UT19]	{rstatic}	●	●		○	●		○					●								●	static analysis	AST visitor
[UT21]	{RTypeInference}	●	●		○	●		○					●		●						○	type inference	AST visitor
[BG20]	{SimilaR}	●	○			○			○				●									plagiarism	PDG, visitor
[Kal+14]	FastR	●	●		○			○	○	○			●									execute R	AST visitor
[R C23]	GNU R	●	●	○	●	○	○	●	○				●									execute R	bytecode
[19]	MRO	●	●	○	○	○																execute R	bytecode
[Nea14]	pqR	●	●	○	●	○	○	●	○				●									execute R	bytecode
[AS12]	Random	●	○	○	○	○				○			○									abstract int.	trace & visitor
[Gar04]	RCC	●	●		●	○	○		○			○	●									transpile to C	CFG, bytecode
[Ber13]	renjin	●	●	○	○	●		○	○			○	●		●							execute R	SSA, CFG
[Flü+20]	Ř	●	●	○	●	●	○	●	○				●	○	●							execute R	SSA, bytecode
[Sen+17]	ROSA	●	●			○			○		○		●		●	○						optimization	visitor
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[Tie23]	{codetools}	●	●	○	○	○	○	○	○				●								●	static analysis	AST visitor
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flowR

github.com/flowr-analysis/flowr

flowR

- A static analysis framework for R

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flowR

- A static analysis framework for R
- Developed here, at Ulm University

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- Let's get back to R:

```r
x <- 4
f <- function() x
body(f) <- quote(y)
y <- 42
f() # 42
```

```r
`if` <- function(...) 42
if(TRUE) print(3) # 42
```

```r
x <- 2
`<-` <- `*`
x <- 21 # 42
```

github.com/flowr-analysis/flowr

flowR

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x <- 2
`<-` <- `*`
x <- 21 # 42
```

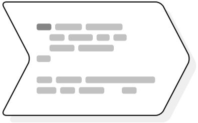
- We have to intertwine dataflow- and control-flow analysis…

github.com/flowr-analysis/flowr

flowR — Architecture

github.com/flowr-analysis/flowr/tree/main/src/dataflow

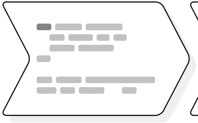
flowR — Architecture



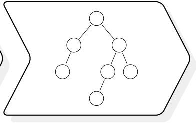
Parse

github.com/flowr-analysis/flowr/tree/main/src/dataflow

flowR — Architecture



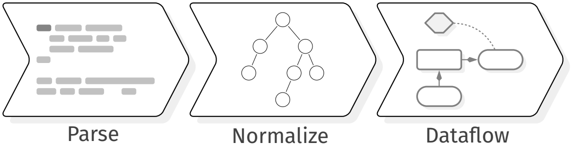
Parse



Normalize

github.com/flowr-analysis/flowr/tree/main/src/dataflow

flowR — Architecture



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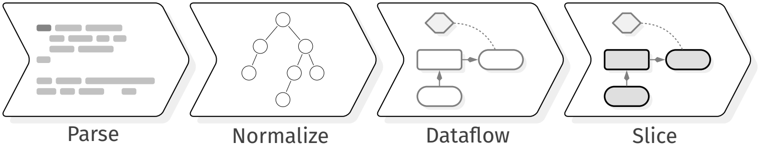
flowR — Architecture



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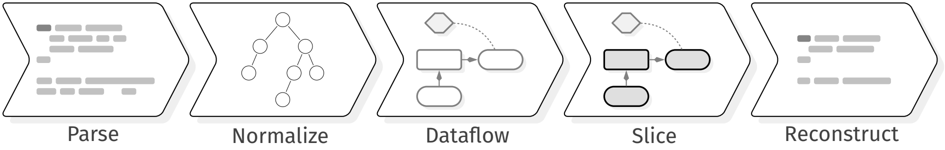

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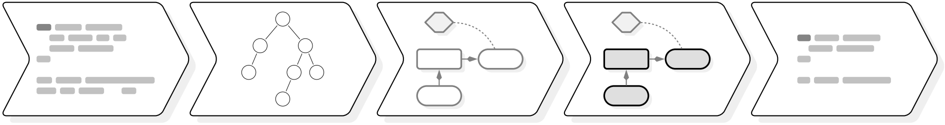


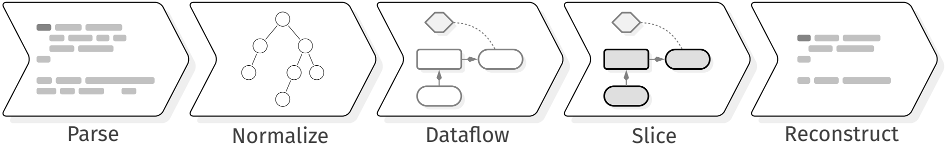

```r
sum  <- 0
prod <- 1
n    <- 10

for (i in 1:(n-1)) {
   sum  <- sum + i
   prod <- prod * i
}

cat("Sum:", sum, "\n")
cat("Product:", prod, "\n")
```

github.com/flowr-analysis/flowr/tree/main/src/dataflow

flowR — Architecture

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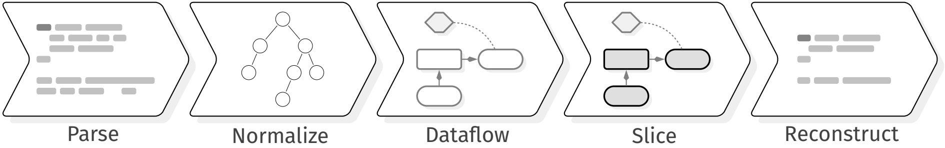


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github.com/flowr-analysis/flowr/tree/main/src/dataflow

flowr — Dataflow

Without Value tracking

flowr — Dataflow

```
x₀ <- 21
y₀ <- 2
z₀ <- x₁ * y₁
```

Without Value tracking

flowr — Dataflow

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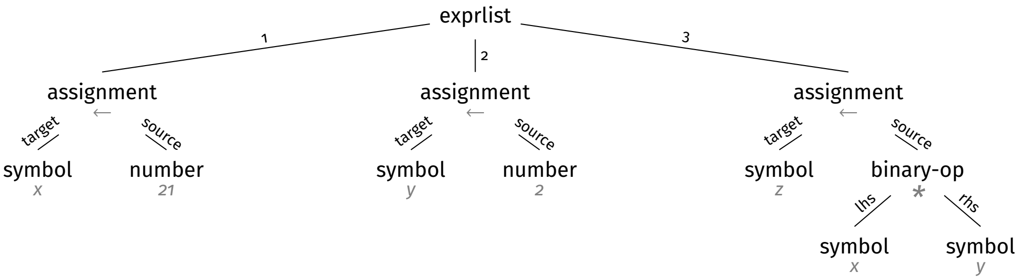
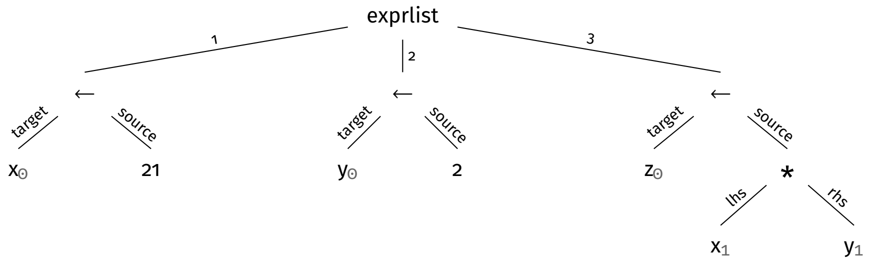


Without Value tracking

flowr — Dataflow

```
x⊙ <- 21
y⊙ <- 2
z⊙ <- x1 * y1
```



Without Value tracking

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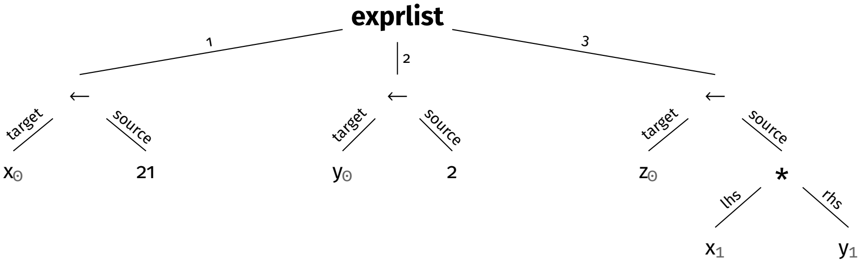

Without Value tracking

flowr — Dataflow

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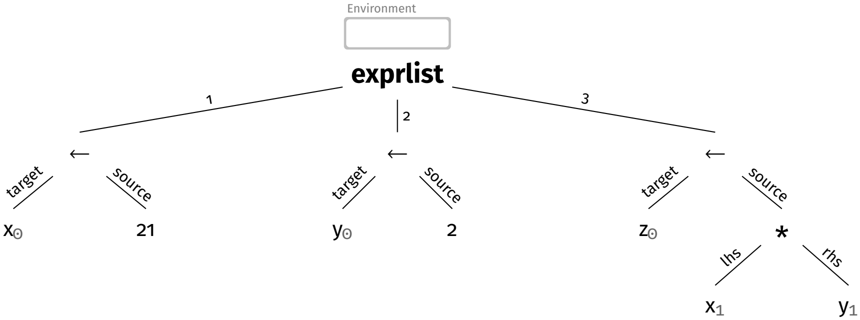

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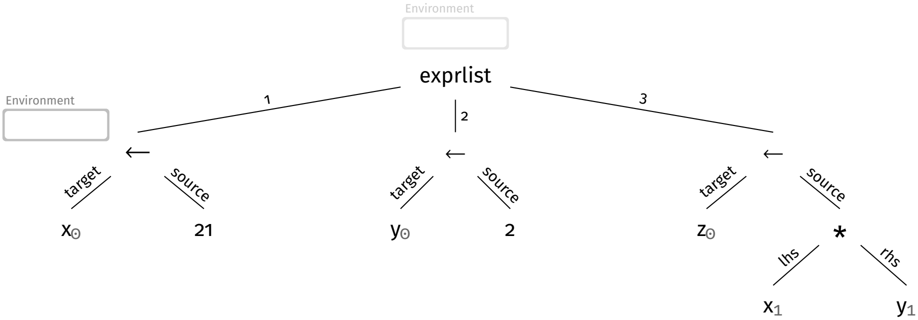

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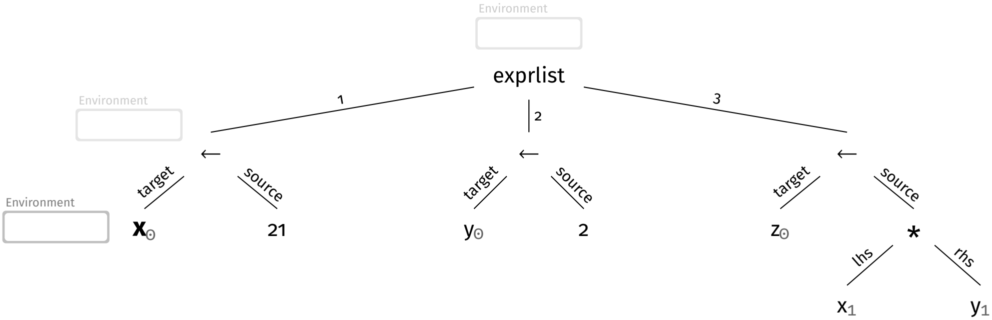

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flowr — Dataflow

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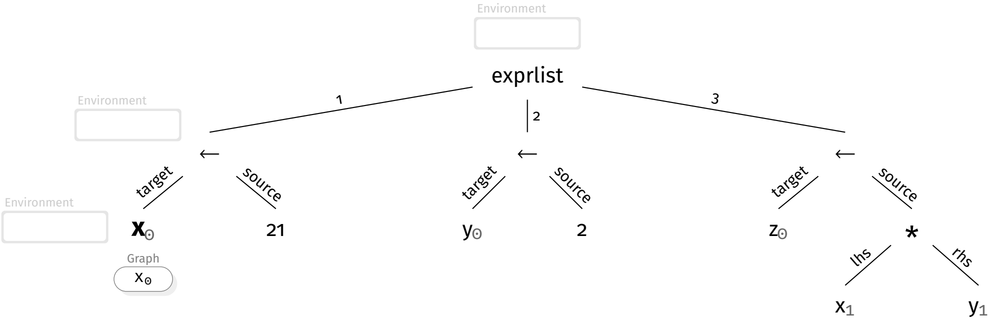

Without Value tracking

flowr — Dataflow

```
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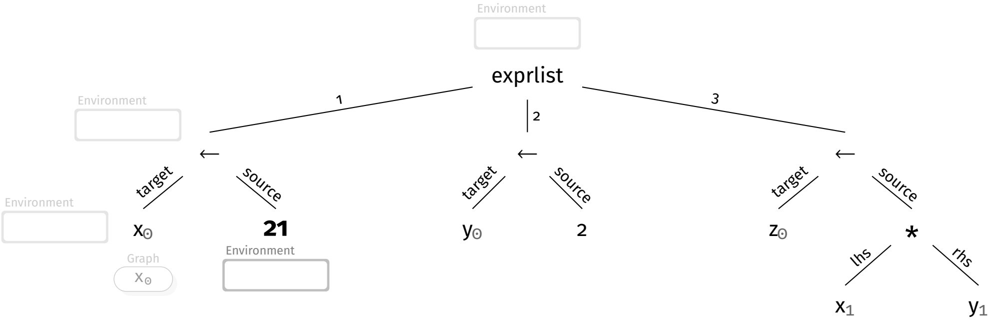
```



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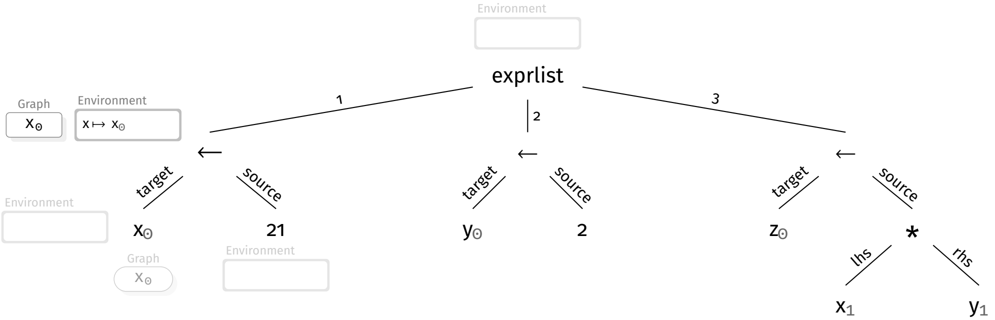
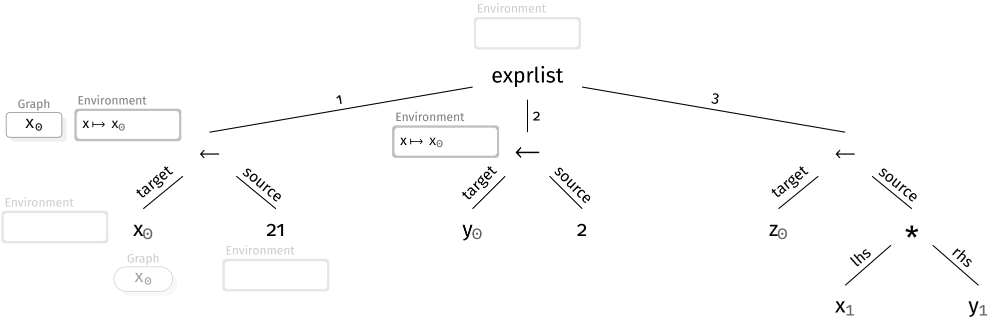


Without Value tracking

flowr — Dataflow

```
x₀ <- 21
y₀ <- 2
z₀ <- x₁ * y₁
```



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flowr — Dataflow

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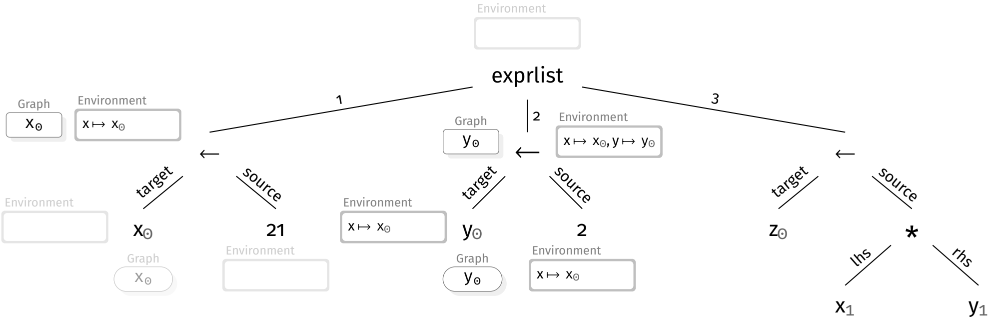
x⊙ <- 21
y⊙ <- 2
z⊙ <- x1 * y1
```



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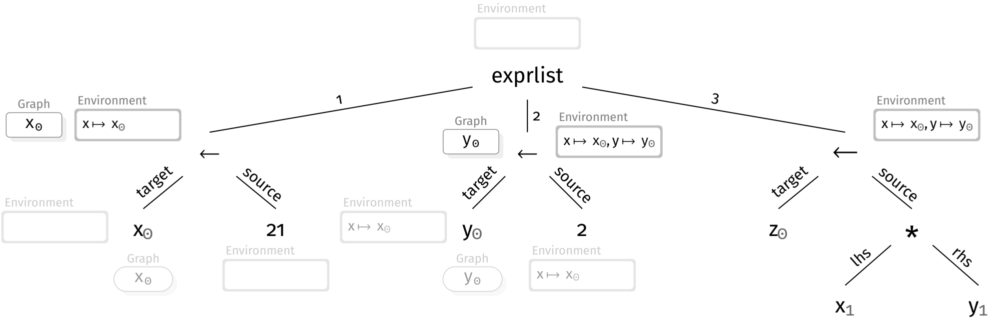

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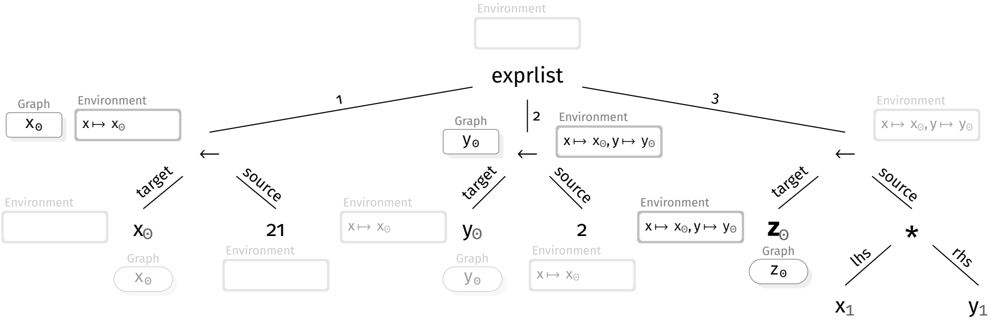

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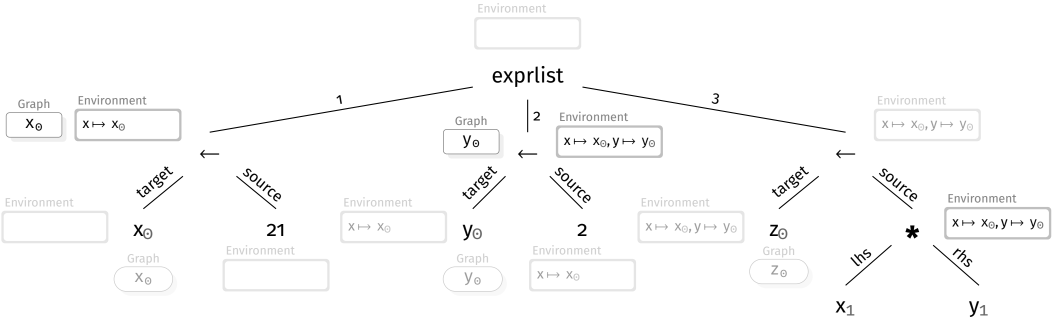

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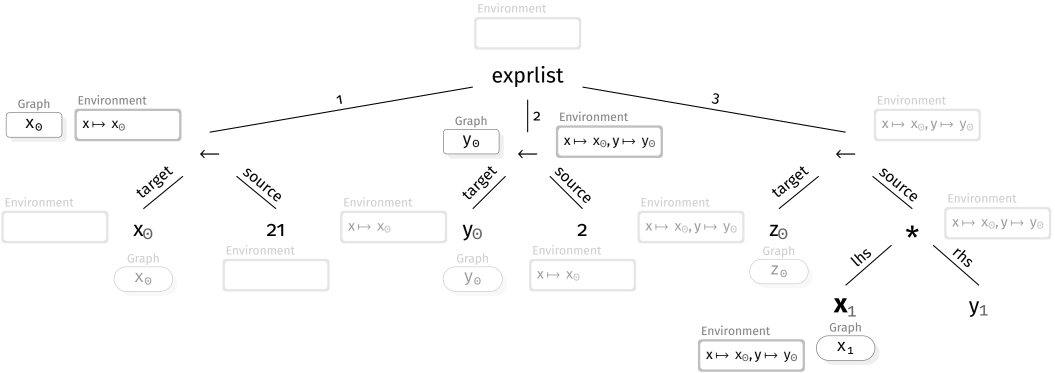

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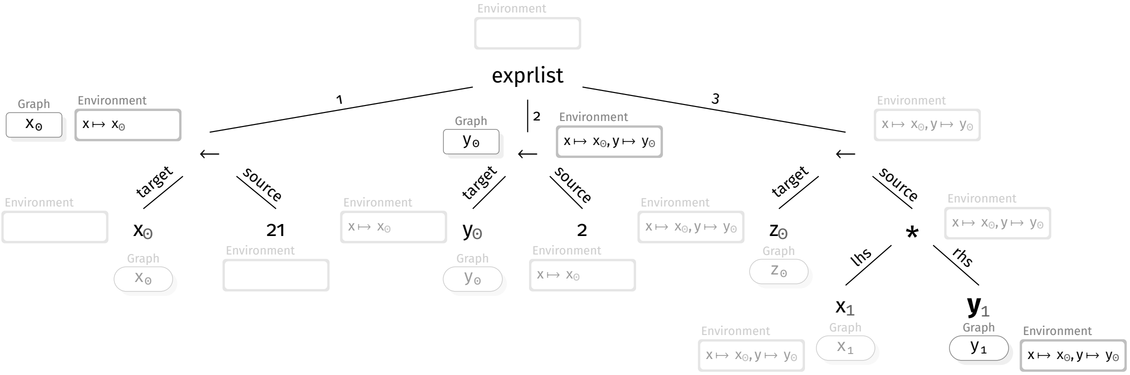

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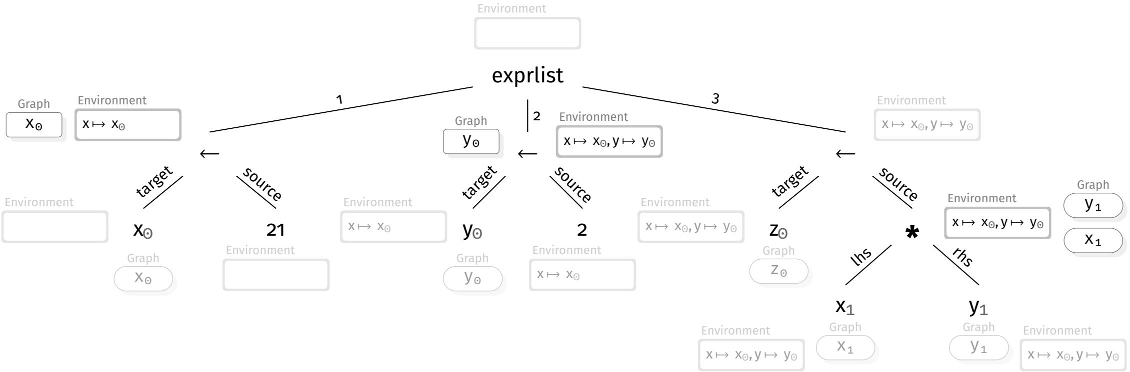

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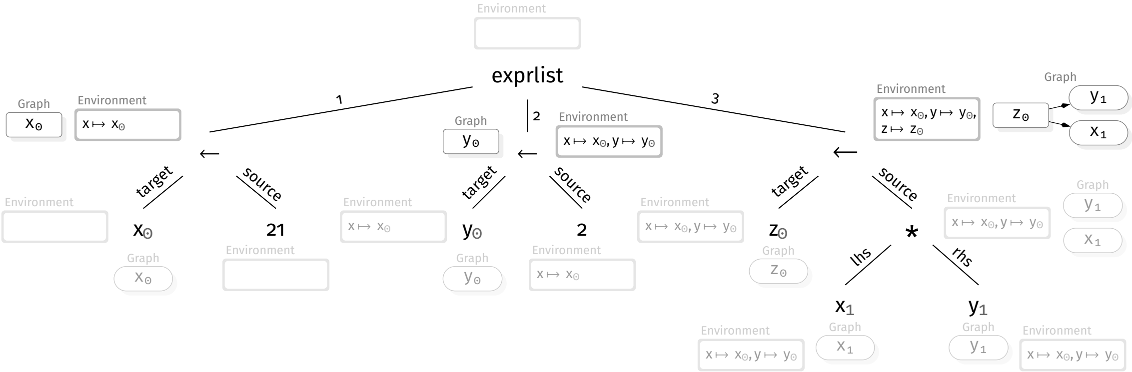
flowr — Dataflow

$x_\odot$ <- *21*
$y_\odot$ <- *2*
$z_\odot$ <- x_1 * y_1



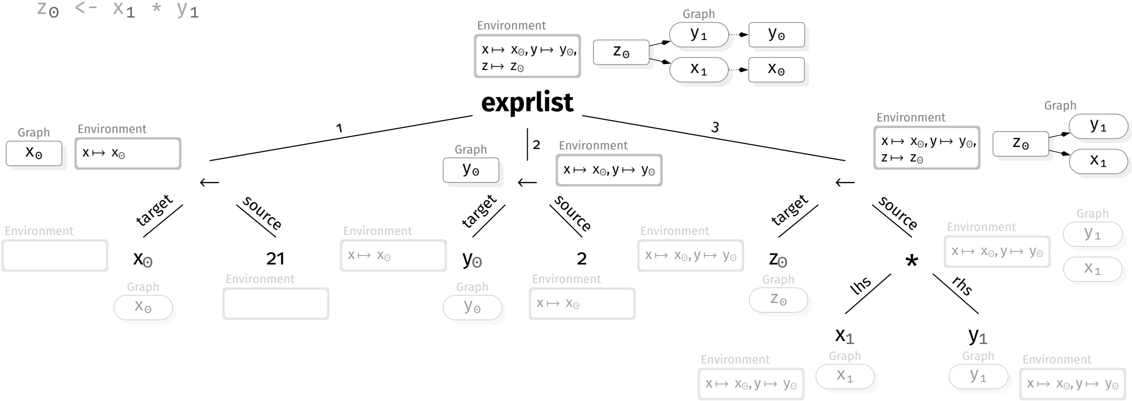
Without Value tracking

flowr — Dataflow

$x_{\odot}$ <- 21

$y_{\odot}$ <- 2

$z_{\odot}$ <- x_{1} * y_{1}



Without Value tracking

flowR — There Is More…

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f₀ <- function(a₀, b₀ = 3) {
   a₁ + b₁
}
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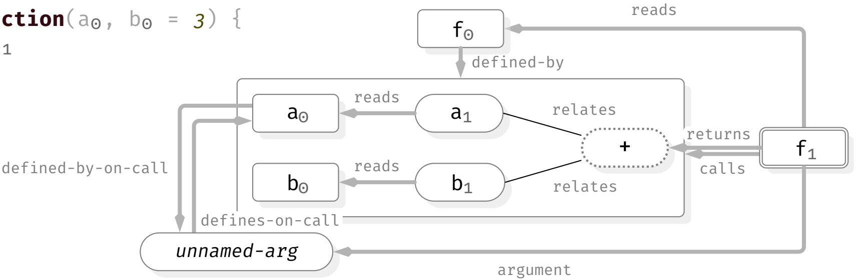

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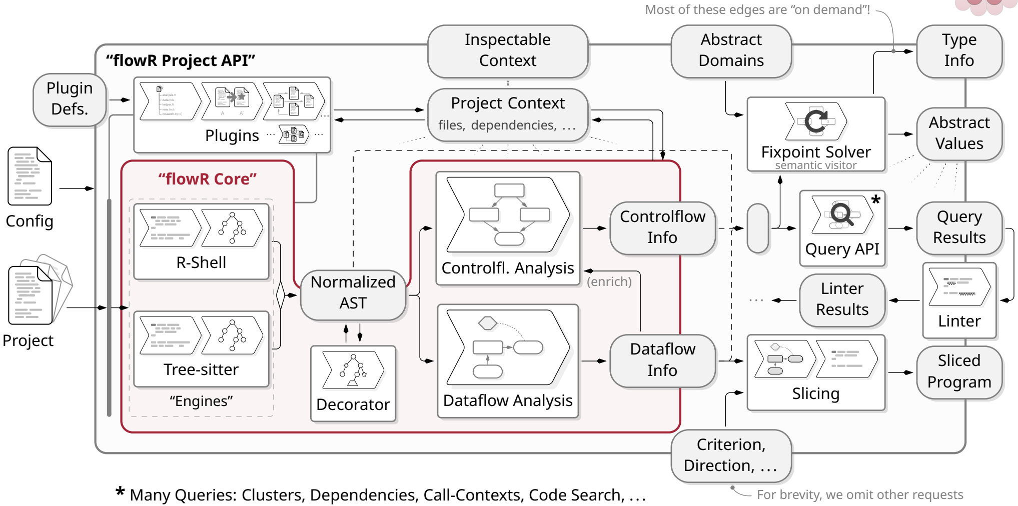

* Many Queries: Clusters, Dependencies, Call-Contexts, Code Search, ...

4. Conclusion

Soundness and Completeness Revisited

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Soundness

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- All properties we derive are true (but we may miss some)
- If we report bugs for violated properties, we produce no false negative

Completeness

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Soundness

- All properties we derive are true (but we may miss some)
- If we report bugs for violated properties, we produce no false negative

Completeness

- We are able to infer all interesting properties in the program
- If we report bugs for violated properties, we produce no false positive

Handling Errors

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- These is still a lot of research required [EN08; Cou21, p. 705]

Oh There Are so Many Tools…

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SootUp
flowR
LiquidHaskell
AbsInt
lintr
Teamscale
Infer
LiSA
CodeQL
sonar
Clang
GCC
ESLint
MOPSA analyzer
frama C

Oh There Are so Many Tools…

LiquidHaskell Verify properties at compile time

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Oh There Are so Many Tools...

Use inference rules for linting ⊢ Infer

Oh There Are so Many Tools…

Clang

Oh There Are so Many Tools…

Determine registers, scheduling, types …
For example:

$$\frac{\Gamma \vdash e_1 : \texttt{int} \quad \Gamma \vdash e_2 : \texttt{int}}{\Gamma \vdash e_1 + e_2 : \texttt{int}}$$

Clang

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Program analysis is important. Additionally AND combined with testing.

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1. How would you capture what a *property* is? ✔
2. How would you phrase that one property is “better” than another? ✔
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4. Why can’t there be a fully automatic, sound, and complete static analyzer for general programs?
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Thanksies!

slides and source code

feedback form

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